Class 7

Biology

Olympiad Program
Volume - II

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6. Reproduction in Plants

Solutions

LEVEL-I

1. Correct option: (D) All the above

There are a number of characteristics that help seeds spread through the air. First of all, because they have wings, they are light and may be easily moved by wind currents. They may now go farther and find new places to germinate thanks to this. Second, the tiny size of these seeds contributes to their capacity to be carried by air. They are easily carried by even light winds due to their compact size, which lowers air resistance. Lastly, some of these seeds have hairy structures, such as tufts or bristles, which further enhance their ability to catch onto air currents and be carried away. Overall, these features collectively contribute to the successful dispersal of seeds through air.

2. Correct option: (C) Both A and B

Different characteristics of seeds that are disseminated through the air help them do so. They can easily be moved by wind currents because they have wings and are light. This enables them to travel farther and arrive at new locations for germination. Second, the size of these seeds is so little that it also helps them travel through the air. Even light breezes can lift them due to their small size, which lowers air resistance. Furthermore, some of these seeds include hairy features like tufts or bristles that improve their capacity to attach to air currents and be blown away. Overall, these characteristics help seeds disperse well through the air.

3. Correct option: (B) Sweet potato

A root that has undergone structural modifications to carry out particular tasks other than anchoring the plant and absorbing water and nutrients is referred to as a modified root. The sweet potato is the right response in the case of the provided possibilities. The sweet potato is a tuber, which is a modified root. To store starch and other nutrients for the plant, it gets bloated and expanded. As a storage organ, the sweet potato differs from a normal root in that it enables the plant to endure adverse conditions and supply energy for growth and reproduction.

The potato is basically a modified stem, more particularly, a tuber, which is an underground stem. The onion is a modified stem, more precisely, a bulbous subterranean stem. Garlic is also a modified stem known as a bulb, much like the onion.

4. Correct option: (A) Spores

By means of spores, bread mould reproduces. Small reproductive cells called spores are created by the mould. These spores are dispersed into the air and are readily transferable to other surfaces, such as additional slices of bread. The spores can germinate and develop into new mould colonies when the right circumstances are present, such as when there is sufficient moisture and food available. This method of reproduction allows bread mould to quickly and efficiently spread and colonize new areas, making it a successful and resilient organism.

5. Correct option: (D) All the above

All of the given options, A) Caltropis, B) Maple, and C) Milkweed, are dispersed by air. Plants frequently disperse their seeds by means of the air to reach new areas. Small seeds produced by Caltropis, also known as Gomphrena globosa, have feathery characteristics that allow them to be carried by the wind over a variety of distances. The characteristic "helicopter" seeds of maple trees rotate as they fall, allowing them to escape the parent tree by catching the wind. In contrast, milkweed produces seeds with silky hairs that

promote wind dispersal. All three of these plants have developed systems to benefit from air currents in order to spread their seeds widely and increase the likelihood that the seeds will find favourable habitats for germination and growth.

6. Correct option: (D) All the above

Animals are responsible for dispersing seeds with hooks, thorns, and hairy portions. As animals move through their surroundings, these adaptations allow seeds to cling to their fur, feathers, or bodies. Burdock and cocklebur seeds, for example, have hooks that allow them to cling to an animal's fur or feathers and travel to new areas. Thorny seeds, such as those in roses or blackberries, can cling to an animal's skin or fur and travel from one place to another. Similarly, seeds from plants with hairy portions, such as dandelions or cattails, can get tangled in an animal's fur or feathers and travel to other locations. Animals are therefore in charge of disseminating all of these different kinds of seeds.

7. Correct option: (C) Both A and B

Both birds and insects act as natural pollinators. For some kinds of plants, especially those with enormous, vividly colored blooms that yield a lot of nectar, birds are known to be efficient pollinators. They frequently have lengthy tongues and beaks that enable them to reach deep inside the blossom to get at the nectar. On the other hand, insects are the most prevalent and effective pollinators. Flowers draw them in because of their vivid hues and alluring aromas. They unintentionally carry pollen from the male reproductive organs (stamens) to the female reproductive organs (pistils) as they go from flower to bloom, aiding in fertilization. Some insects, such as bees, butterflies, and beetles, are particularly important pollinators due to their ability to carry large amounts of pollen on their bodies. Overall, both birds and insects play crucial roles in the natural process of pollination, ensuring the reproduction and survival of many plant species.

8. Correct option: (D) Bryophyllum

Since bryophyllum is a special plant that reproduces through buds on its leaves, it is the right response. It is possible for a plant to create new progeny by a technique known as vegetative propagation that does not require seeds or blooms. Along the edges of the leaves, the buds, also known as plantlets, form before falling off, whereupon they take root and develop into new plants. Since it enables quick and effective multiplication, this type of reproduction is beneficial for Bryophyllum because it ensures the survival and expansion of the species.

Rhizomes are used in the rhizomes of sugarcane, a long, perennial grass. The seeds of the biennial turnip are used to spread the plant. Its tubers are used to spread the herbaceous perennial plant known as the potato.

9. Correct option: (B) Balsam

Impatiens spp., also known as balsam plants, are noted for their explosive seed dissemination technique. Balsam ripe seed pods quickly break open when touched or otherwise disturbed because of the strain that the walls of the pods have built up from drying. The result is a forced ejection of the seeds from the pod, which frequently disperse far from the parent plant. By being projected away from the parent plant, these seeds have a better chance of locating places where they may grow and gerinate. This type of seed dispersal aids the balsam plant in colonizing new areas.

The other options (A) Coconut, (C) Mango, and (D) Tamarind do not disperse seeds by bursting of fruits; they have different modes of seed dispersal.

10. Correct option: (B) Entomophily

The term "entomophily" describes flower pollination that is aided by insects. As they gather nectar, insects like bees, butterflies, moths, and beetles are essential in moving pollen from one blossom to another. Bright colors, alluring fragrances, and nectar cues

are common characteristics of flowers suited for entomophily to entice insects. Numerous coevolutionary adaptations between plants and their insect pollinators have resulted from this type of ubiquitous pollination method in the plant kingdom.

Pollination by creatures other than insects, such as bats, birds, and rodents, is known as zoophily. Pollination by snails and slugs is known as malacophily. Ornithophily is the practice of using birds to pollinate plants.

11. Correct option: (D) Ovule

The ovule in flowering plants transforms into a seed during fertilization. The female gametophyte and egg cell are located in the ovule, a structure in the ovary of the flower. When a pollen tube transports a pollen grain to the ovule, the male and female gametes can unite to produce fertilization. A zygote is created as a result of this fusion, and it grows into an embryo inside the ovule. The ovule undergoes modifications as the embryo grows, ultimately becoming a seed. The embryo, a food source for the growing plant (in certain cases endosperm), and a protective seed coat are all parts of the seed.

12. Correct option: (D) All the above

Adapted for wind dispersal, seeds with hairy structures use their hairs or tiny fibers to catch the wind and move away from the parent plant. In the case of cotton, the hairs help spread seeds and the fibers are well-known for their usage in textiles. Another plant with hairy seed structures that can be carried by air currents is called madar (Calotropis). Dandelion seeds, on the other hand, contain feathery hairs that form a "parachute," enabling the seeds to be lifted by even the smallest wind. The existence of these hairy appendages in cotton, madar, and dandelion seeds demonstrates their wind dispersal adaptations, increasing the likelihood that the seeds would land in favorable conditions for germination and growth.

13. Correct option: (D) All the above

The correct answer is D) All the above. This is so because each of the pairs is matched to a particular component of the structure of a flower. A) The term "calyx" describes the outermost whorl of a flower made up of sepals, which are typically green and serve to shield the budding bud. B) The corolla, or second whorl, is made up of petals that are frequently colorful and appealing, assisting in pollination. C) The third whorl, which contains stamens, the male reproductive organs that create pollen, is where an androecium is located. As a result, each pair closely matches a distinct part of a flower's anatomy, including its defense, appeal, and reproductive functions.

14. Correct option: (D) All the above

The answer is D) All the above. A flower is said to be complete when it has all four major components: sepals, petals, stamens, and pistils. Hibiscus, Datura, and Ipomea are a few examples of plants with full blooms. The flower's outermost structure, called sepals, is typically green and has a protective function. The flower's vibrant and enticing petals are what frequently draws pollinators. While pistils are the female reproductive organs that house the ovary and receive pollen for fertilization, stamens are the male reproductive organs that create pollen. All three of the aforementioned plants are regarded as full flowers because they include all four of these components.

15. Correct option: (D) All the above.

The female reproductive organs of a flower receive pollen from the male reproductive organs through pollinating agents, which allows fertilization and the development of seeds. Animals, air, and water are a few examples of these agents. Pollen grains are spread onto plant female reproductive organs through airborne pollination, which is caused by wind currents. Pollen is transported across water to reach the female reproductive organs in aquatic plants, which pollinate less frequently. The most common

and effective mode of pollination is by animals, with bees, butterflies, birds, and bats playing a critical part in the movement of pollen between flowers. Therefore, the correct answer is D) All the above, as all three pollinating agents - air, water, and animals - contribute to the pollination process in different ways.

16. Correct option: (B) Madar

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17. Correct option: (A) Fleming

The answer is A) Fleming, referring to Sir Alexander Fleming. In 1928, he is credited with discovering penicillin. Fleming discovered a mould called Penicillium notatum generated a chemical that killed a variety of bacteria when he was working at St. Mary's Hospital in London. As a result of this unintentional discovery, antibiotics were created, changing medicine by effectively treating bacterial illnesses. The discovery of penicillin as a potent and well-known antibiotic was made possible by Fleming's observation of the mold's antibacterial characteristics.

18. Correct option: (B) Ovary

The correct answer is B) Ovary. After successful fertilization, the ovary, which is a component of the flower's reproductive structure, turns into a fruit. The female reproductive cells are housed in one or more of its ovules. The fertilization process takes place inside the ovary after a flower has been pollinated, often by the transfer of pollen from the stamen to the stigma. The ovary undergoes a transformation into a fruit as a result of this fertilization, which can serve a variety of purposes, including preserving and distributing the seeds produced by the fertilized ovules. By causing the growth of fruits, the ovary thus plays a critical part in the reproductive cycle of flowering plants.

19. Correct option: (B) Budding

The correct answer is B) Budding. Budding is the process through which yeast cells divide. A little protrusion or bud develops on the parent yeast cell's surface during budding. This bud becomes bigger over time while still being connected to its parent cell. The bud eventually splits off from the parent cell and develops into a distinct yeast cell. By using this process, yeast cells can create progeny that are genetically identical to themselves. Asexual reproduction known as budding is typical of single-celled organisms like yeast.

20. Correct option: (C) Thalamus

The correct answer is C) Thalamus. The inflated base of the flower, often referred to as the thalamus or receptacle, is where all of the floral organs are whorl-mounted. It gives the flower's numerous parts, such as the sepals, petals, stamens, and carpels, support and structure (which include the ovary). The reproductive and defensive organs of the flower are positioned on the thalamus, which facilitates their coordinated growth and operation. It is a crucial component of the flower's anatomy, assisting in the appropriate positioning and display of the floral parts to maximize their capacity for reproduction.

21. Correct Matching is: a-R, b-P, c-Q, d-T, e-S

a) Thalamus - R) Seat of a flower:

The bulging base of a flower, sometimes referred to as the receptacle or thalamus, is where all the floral organs are attached. It acts as a base or "seat" for the arrangement of the other sections, offering support and structure for the flower's constituent parts.

b) Calyx - P) Protects the parts of a flower:

The flower's outermost whorl, the calyx, is made up of sepals. It frequently offers support while also enclosing and guarding the developing inner flower elements. The fragile reproductive components within the flower can be protected by sepals.

c) Corolla - Q) Attracts the insects:

The second whorl of the flower, which is made up of petals, is called the corolla. The flower's petals, which are frequently vividly colored and come in a variety of forms, draw insects and other pollinators. It is essential for pollinators to be attracted to the corolla in order to promote pollination.

d) Androecium - T) Filament with bulb-like pollen sac

The stamens, which are the male reproductive components of the flower, are referred to as the androecium. A filament (stalk) and an anther (pollen sac), which is where pollen is made, make up each stamen. The pollen-containing anther, which is supported by the filament, occasionally has a bulb-like form.

e) Gynoecium - S) Pistil

The flower's female reproductive system, which may include one or more carpels, is represented by the gynoecium. The ovary, style, and stigma are located inside the carpel. The term "pistil" is frequently used synonymously with "carpel," and it refers to the female reproductive system, which includes the stigma, which receives pollen, the style, which supports the stigma, and the ovary, which houses the ovules.

22. Correct Matching is: a-T, b-S, c-Q, d-R, e-P

a) Complete flower - T) A flower that has all the four whorls

Sepals, petals, stamens, and pistils are the four major whorls of floral organs that make up a full flower. These are the fundamental elements of a flower that work together to support its reproduction process.

b) Incomplete flower - S) A flower in which any of the four whorls is missing

One or more of the four primary whorls are absent in a type of flower known as an incomplete flower. This indicates that one or more of the flower's structural components—sepals, petals, stamens, or pistils—is absent.

c) Unisexual flower - Q) A flower that has either stamens or pistil

A flower that is missing one or more of its four primary whorls is said to be incomplete. This denotes a structural flaw in the flower, such as the absence of sepals, petals, stamens, or pistils.

d) Bisexual flower - R) A flower that has both stamens and pistil

A bisexual flower, often known as a perfect flower, has both female (pistil) and male (stamen) reproductive organs. This enables the flower to carry out both male and female reproductive processes, enabling cross-pollination and self-pollination.

e) Anther - P) Pollen sac

The area of the stamen where pollen is produced called the anther. Pollen grains are held in structures resembling sacs called pollen sacs. The male gametes required for fertilization are contained in these grains.

23. Correct Matching is: a-T, b-S, c-Q, d-R, e-P

a) Pollination - T) Transfer of pollen grains from anther to stigma

Pollination is the process of moving pollen grains from a flower's male reproductive structure, or anther, to its female reproductive structure, or another flower, called a stigma. The success of fertilization depends on this transmission.

b) Fertilization - S) Fusion of male and female gametes

The process through which the male and female gametes (sperm and egg cells) join forces to create a zygote is called fertilization. The plant's subsequent generation is starting with this zygote.

c) Zygote - Q) Formed after the fertilization process

The cell that results from the union of the male and female gametes during fertilization is known as the zygote. It is the first cell of an embryo, which then develops into a new individual plant.

d) Vegetative reproduction - R) Reproduction by somatic parts of a plant

In asexual plant reproduction known as vegetative reproduction, non-reproductive plant elements like stems, roots, or leaves are used to create new individuals. Neither seeds nor spores are produced during this process.

e) Spores - P) Asexual reproduction of fungus

A few plants and fungi develop reproductive structures called spores. They are microscopic cells that, in the right circumstances, can grow into new individuals, enabling asexual reproduction in organisms like fungus.

24. Correct Matching is: a-R, b-P, c-S, d-Q

	Group A	**Group B**
a)	Pollen grain	R) Anther
b)	Ovule	P) Ovary
c)	Reproduction through eyes	S) Potato
d)	Reproduction through leaves	Q) Bryophyllum

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LEVEL - II

1. Correct option: (B) Formation of seed without syngamy and meiosis.

The term "apomixis" describes the generation of seeds without the meiosis and syngamy processes. A zygote is created through the union of gametes, which is known as syngamy. On the other hand, meiosis is a type of cell division where the number of chromosomes is cut in half. In the event of apomixis, neither of these two critical processes is involved in the formation of the seeds. Instead, without fertilization, the seeds are created by the direct growth of the egg cell. As there has been no genetic recombination or variation introduced, this indicates that the offspring generated through apomixis are genetically identical to the parent plant. As a result, option B is the right response because it explains apomixis in an exact manner.

2. Correct option: (C) Papaveraceae

The Papaveraceae family of plants have the censar mechanism of seed dispersion. When touched or agitated, the fruit capsule's unique dispersal mechanism causes the seeds to eject, frequently as a result of the fruit walls' drying up or withering. This process in Papaveraceae plants—including poppies—projects seed out of the fruit capsule quickly and frequently violently, facilitating their dispersal across a larger region. By lowering competition with parent plants and assisting in seed distribution, this adaptation encourages the colonization of new areas.

3. Correct option: (D) Chara

Chara is a monoecious plant. On the same individual plant, monoecious plants have both male and female reproductive organs. Green algae known as Chara are typically found in freshwater habitats. The male reproductive structures (antheridia) and female reproductive structures (oogonia) are found on the same thallus in monoecious Chara species (plant body). Due to the proximity of both types of reproductive organs on the same plant, this configuration permits the possibility of self-fertilization.

4. Correct option: (A) Carrion flies

True, carrion insects are mostly responsible for pollination in Rafflesia. One of the largest individual flowers in the world are produced by the parasitic flowering plant genus Rafflesia. These blooms give off an unpleasant smell that is similar to that of rotting flesh, luring carrion flies. The flies are drawn to the flower because they believe it to be a food source, and while they are searching for food, they unintentionally spread pollen between the male and female flowers of various plants, aiding in pollination. Rafflesia's potent aroma and unusual pollination method are adaptations to its particular ecological niche.

5. Correct option: (B) 8

The fruit fly, Drosophila melanogaster, has eight chromosomes in each meiocyte (a cell undergoing meiosis). Fruit flies contain two sets of chromosomes since they are diploid organisms. However, the number of chromosomes is decreased by half during meiosis, the cell division process that results in the creation of gametes (sperm and egg cells). When it comes to fruit flies, the reduction division creates four haploid cells with eight chromosomes apiece. This haploid number of chromosomes guarantees that the subsequent diploid zygote will have the typical number of chromosomes for the fruit fly species, which is 16, when fertilization takes place and the sperm and egg unite.

6. Correct option: (A) Great synchrony between sexes

Externally fertilized organisms, in which the union of the gametes takes place outside the body, demonstrate excellent sex-to-sex synchronicity. Both males and females discharge their gametes (sperm and eggs) into the environment during external fertilization, frequently in close temporal and geographical proximity. In aquatic creatures, the gametes must meet and fuse in the water before they scatter or are eaten, therefore this

synchronization is essential for proper fertilization. Coordination of timing and position during external fertilization is necessary to increase the likelihood that gametes will come into contact during successful reproduction. Internal fertilization, which is common among terrestrial animals, allows for greater gamete protection from external variables and reduces the requirement for synchronized release in the short term.

7. Correct option: (B) Multicarpellary syncarpous

The papaver has a multicarpellary syncarpous pistil, which is the plant's female reproductive system. As a result, the pistil is made up of many carpels that have been united into a single structure. The separate carpels of papaver combine to create a syncarpous pistil. The stigma, which is where pollen lands, the style, which is a stalk connecting the stigma to the ovary, and the ovary normally make up each carpel (containing ovules that will develop into seeds after fertilization). In many flowering plants, the fusing of carpels into a syncarpous pistil is a common structural adaptation. In papaver, it makes reproductive activities like pollination and seed development easier.

8. Correct option: (D) Continue the species forever

A crucial biological process that allows living things to perpetuate their species is reproduction. By reproducing, organisms create children who inherit their genetic code, preserving genetic diversity and allowing for environmental adaption. While an organism's ability to develop, use its energy, and maintain its various organs is necessary for its survival and well-being, reproduction performs the special function of transferring genetic features to succeeding generations. Because of their ability to adapt to changing environmental conditions and evolve over time, animals are able to have long lives on Earth.

9. Correct option: (C) 2 polar nuclei + 1 male gamete nucleus

The fusing of two polar nuclei and one male gamete nucleus results in the formation of the main endosperm nucleus (PEN). When flowering plants undergo twofold fertilization, this procedure takes place. Two male gametes are released when a pollen tube enters the ovule; one combines with the egg cell to make the zygote, and the other unites with the two polar nuclei in the embryo sac's center cell to form the PEN. The endosperm, a nutrient-rich tissue that feeds the growing embryo, is created from the primary endosperm nucleus.

10. Correct option: (D) Tapetum nourishes the developing pollen

A specialized layer of cells called the tapetum is present in the anther of a flower. Its primary purpose is to nourish and nurture the growing pollen grains. It secretes a variety of compounds, including proteins, lipids, and enzymes, which are crucial for the pollen's healthy growth and development. The pollen grains would not be able to grow and perform effectively for pollination without the tapetum because they would not be provided with the essential nutrients. Therefore, by guaranteeing the successful production of pollen grains, the tapetum plays a critical part in flowering plants' reproductive process.

Exine refers to the pollen's tough outer covering. The tissue from which microspores are produced is known as sporogenous tissue. The endothecium is the innermost layer of the anther wall, and it does not produce the microspores.

11. Correct option: (A) allogamy

In the real genetic meaning, cross-pollination is the act of moving pollen from one plant's reproductive organs, like stamens, to another plant's reproductive organs, like pistils, within the same species. This could happen because of the wind, insects, birds, or other forces. The term "allogamy," denoted by option A, especially alludes to cross-pollination, in which pollen from the blooms of one plant fertilises the ovules of another plant. As a

result of this genetic exchange, a species' genetic diversity is encouraged, which can improve adaptation and evolutionary success.
Autogamy (B) is self-pollination, where pollen from the same plant reaches its own stigma. Cleistogamy (C) is the production of closed, self-fertilizing flowers. Geitonogamy (D) is a subset of allogamy, involving pollination between flowers of different individuals within the same population. Among these options, allogamy (A) best captures the essence of true cross-pollination, fostering genetic variability.

12. Correct option: (C) Apogamy

In the fern life cycle, the prothallus—a tiny, heart-shaped structure—can develop into a new plant without the requirement for fertilization. This process is known as apogamy. This process involves the direct formation of a sporophyte plant from a gametophyte (the prothallus) without the fusion of gametes (sperm and egg cells). This avoids the typical meiosis and fertilization stages of sexual reproduction. Due to the fact that asexual reproduction does not entail the recombination of genetic material from two separate sources, it is possible to produce sporophytes that are genetically identical to their parent plants. In some fern species, this may support clonal reproduction and genetic stability.
Apospory is the formation of a gametophyte (n) directly from a cell of the sporophyte (2n) without meiosis. Parthenocarpy is the development of fruits without fertilization. Organogenesis is the process of forming organs.

13. Correct option: (A) 1 diploid and 1 triploid cell

One diploid cell and one triploid cell make up a mature ovule following double fertilization, which happens in angiosperms. The procedure entails the merging of two sperm cells with two distinct nuclei inside the ovule's embryo sac. The embryo is created when one sperm cell combines with an egg cell to create a diploid zygote. A triploid cell known as the major endosperm nucleus is created when the other sperm cell merges with two polar nuclei. The endosperm, a nutrient-rich tissue that sustains the growing embryo, will develop from this triploid nucleus. Due to this special property of twofold fertilization, the embryo and endosperm within the seed develop in a coordinated manner, which helps flowering plants to carry out their reproductive strategy.

14. Correct option: (B) Sucker

Plants that reproduce vegetatively by a process known as "suckering" include elm (Ulmus), dandelion (Taraxacum), and different rose species. Suckering is the process through which new shoots or stems develop from the parent plant's roots. These "sucker" shoots, which are also known as lateral buds, develop from the plant's subterranean parts, usually the roots or rootstock. These suckers produce new plants that are genetically similar to the parent plant as they mature into independent shoots. This kind of vegetative reproduction aids in the growth of plant populations and the formation of clonal colonies, which are collections of plants that are virtually identical clones of one another and descended from the same parent.

15. Correct option: (B) small, producing a large number of dry pollen grains.

Flowers that are pollinated by the wind often have a tiny size and produce a lot of dry pollen grains. These flowers lack the eye-catching hues and features that draw pollinators from animals. The efficient generation and spread of pollen, which is frequently tiny, light, and non-sticky, is instead given top priority. The huge amount of pollen grains thrown into the air makes up for the absence of nectar and aesthetic features. The tremendous pollen production and the pollen grains' light weight boost the likelihood of successful wind pollination because wind is these plants' major method of moving pollen from one flower to another. Wind-pollinated plants can improve their reproductive efficiency thanks to this adaptation without the need for animal pollination.

16. Correct option: (C) Morphology, vigour and disease resistance

Morphology, vigour, and disease resistance are the traits that carry over from generation to generation in vegetative propagation by tuber. This means that each new generation of tubers grown using this technique will maintain the same physical characteristics, pace of growth, and disease resistance. Farmers and gardeners benefit from this since they can rely on these characteristics to be present in the progeny, assuring a reliable and appealing harvest or plant. Farmers may guarantee a high-quality output and lower the chance of crop loss due to illnesses by choosing tubers with the necessary morphology, vigour, and disease resistance.

17. Correct option: (D) Pollen and endosperm

The process known as "Xenia" occurs when the endosperm, a nutritive component that surrounds the embryo inside a seed, exhibits traits that are influenced by the maternal parent. Metaxenia, on the other hand, describes how the paternal parent affects the properties of both the endosperm and the embryo. These phrases explicitly refer to the contributions made by the mother (endosperm) and father (pollen) in the formation of seeds. Endosperm feeds the developing embryo while pollen carries the genetic material that joins with the egg cell to create the embryo. Xenia and metaxenia effects are seen in the traits expressed by the endosperm and sometimes the embryo, which can be influenced by the genetic makeup of the pollen donor, thus highlighting the contribution of both pollen and endosperm to the characteristics of the seed.

18. Correct option: (C) The flowers pollinated by the flies and beetles secrete foul odor to attract them

The statement that is not true is C) "The flowers pollinated by the flies and beetles secrete foul odor to attract them." In actuality, flowers that are pollinated by beetles and flies usually have strong, enticing scents or even imitate the aroma of decomposing stuff. Because they deposit their eggs in conditions that resemble rotting materials, flies and beetles are frequently drawn to those smells. By producing scents that are similar to these ones, flowers have evolved to take advantage of this behaviour and draw these pollinators. This adaptation emphasizes the variety of tactics used by plants to draw various pollinators, each one catered to the distinct traits and behaviors of the pollinating insects.

19. Correct option: (B) Sporopollenin

Due in large part to the fact that sporopollenin makes up their exine, pollen grains have the amazing capacity to withstand high temperature changes and desiccation. The pollen grain's outer layer of protection is made of the strong and chemically resistant biopolymer known as sporopollenin. By providing protection from extremes in temperature, UV radiation, and moisture loss, this special chemical protects pollen from environmental challenges. In order to successfully pollinate and fertilize flowers, pollen grains must be able to preserve their viability and integrity along the travel from the anther to the stigma of the flower. The inclusion of sporopollenin in the exine is an essential trait that helps plants reproduce and survive.

20. Correct option: (C) It can propagate vegetatively at a phenomenal rate

The main factor that makes water hyacinth one of the most invasive weeds is its exceptional capacity for vegetative propagation at an incredibly quick rate. The Eichhornia crassipes, sometimes known as the water hyacinth, is distinguished by its aggressive growth and reproductive strategy, which involves the development of multiple daughter plants via vegetative mechanisms like stolons and runners. These daughter plants can quickly encircle bodies of water, generating dense mats that obstruct sunlight, water flow, and oxygen exchange, upsetting aquatic ecosystems. Water hyacinth outcompete native plants and swiftly colonize and take over aquatic settings. This results in serious

ecological and economic issues, including decreased biodiversity, altered water quality, and obstructions to navigation and leisure activities.

21. Correct option: (D) Monocystis

An instance of sexual reproduction known as isogamy occurs when the gametes are morphologically identical, or the same size and shape. Many unicellular creatures, like the parasitic flatworm Monocystis, exhibit isogamy. The Coccidia class of parasitic protozoa includes the genus Monocystis. It is distinguished by having an isogamous reproduction mode in which a zygote is produced by the union of two isogametes that are structurally similar in size and appearance. This is distinct from anisogamy, which occurs in many other organisms and involves the fusion of gametes of different sizes. As a part of its life cycle, Monocystis reproduces isogamously in the reproductive organs of earthworms. Malaria is brought on by the multicellular parasite Plasmodium. A flatworm called a planaria reproduces through fission. Hydra is a freshwater polyp that reproduces by budding.

22. Correct option: (C) Perennial plants

It is true that perennial plants do not go through clean-cut vegetative, reproductive, and senescent periods. Long lifespans and the absence of clearly defined phases, which are present in annual and biennial plants, are characteristics of perennial plants. Annual plants evolve from vegetative growth through reproduction and senescence during the span of one year. Contrarily, biennial plants have a two-year life cycle, with the first year being devoted to vegetative growth and the second year being characterised by reproduction and eventual senescence. Perennial plants, on the other hand, lack such distinct periods. Over a longer period of time, they continue to develop, reproduce, and age in an unpredictable way, frequently alternating between vegetative and reproductive growth. This flexibility in growth and reproductive patterns is a characteristic feature of perennial plants.

23. Correct option: (B) Pre-fertilization, fertilization, post fertilization

B) Pre-fertilization, fertilisation, and post-fertilization is the correct response. The three primary stages of sexual reproduction are as follows. Pre-fertilization, the initial phase, sees the production and development of gametes (sperm and egg). Processes like gametogenesis, in which gametes are created by meiosis, are included in this stage. The sperm and egg combine to produce a zygote during the second stage, which is called fertilization. Because it mixes the genetic makeup of both parents, this period is critical. Post-fertilization involves the zygote's growth and development into an embryo, and it is the last step. Cell division, differentiation, and the development of diverse tissues and organs are all part of this period. The continuance of the species through sexual reproduction depends on these three stages overall.

24. Correct option: (A) Node

Nodes are essential for producing new plants during vegetative propagation. Nodes are certain locations along a stem from which buds, branches, or leaves grow. Nodes have the power to form roots and produce new plants when they come into touch with moist soil or water. Meristematic tissues, which are in charge of cell division and proliferation within nodes, support this process. A new plant begins to grow once the node establishes contact with an appropriate medium, going through physiological changes that allow for a type of asexual reproduction where genetically identical children are created from a parent plant.

The other options are incorrect. A shoot is a young, growing plant. An axial shoot is a shoot that grows from the main stem of a plant. A seed is a reproductive structure that contains an embryo plant.

25. Correct option: (C) Cellulose and pectin

The correct answer is C) Cellulose and pectin. The inner layer of the pollen grain's wall is called the pollen intine, and it is mainly made of cellulose and pectin. The pollen grain and other plant components, including cells, are structurally supported by cellulose, a complex carbohydrate. Pectin is another polysaccharide that is involved in cell adhesion and serves a role in preserving the integrity of the cell wall. The pollen grain is protected and is able to maintain its structure while it is subjected to a variety of environmental stressors during pollination and fertilization thanks to the strength and elasticity that cellulose and pectin together provide to the pollen intine. Pollen grains may successfully carry male gametes to female reproductive organs in plants thanks to their composition.

The pollen grain's exine is its outer coating. It is formed of sporopollenin, a very resilient substance that aids in shielding the pollen grain from the outside elements. The cell membranes of both plants and animals contain a particular form of steroid called ergsterol. Pollen grains don't contain any of it. Although they are absent from pollen grains, phospholipids and cholesterol are also present in cell membranes.

26. Correct option: (A) Vascular cambium is absent

Grafting is the process of uniting plant tissues from two separate plants so that they develop as a single organism. Grafting is possible in dicot plants because they include vascular cambium, a layer of meristematic tissue that promotes lateral growth and the development of secondary vascular tissues. These tissues enable the integration of the circulatory networks of the two plants, preserving the flow of nutrients and water. However, vascular cambium is missing in monocot plants like grasses and lilies, and their vascular bundles are organized in a dispersed way as opposed to creating the distinct rings found in dicots. It is difficult to successfully graft monocots due to the lack of vascular cambium and the different vascular arrangement because there is insufficient tissue to form a vascular connection between the scion (upper part of the graft) and the rootstock (lower part of the graft), which ultimately reduces graft compatibility and success.

The other options are incorrect. Meristematic tissue is present in both monocots and dicots. Collateral open vascular bundles are found in both monocots and dicots. Radial vascular bundles are only found in dicots.

27. Correct option: (D) Synergids, egg cell, micropylar

A flowering plant's ovule contains a specific structure called the egg apparatus at one end of the embryo sac. One egg cell and two synergids make up this organism. Supportive cells called synergids help the pollen tube go toward the embryo sac during fertilization. The zygote, or first cell of the new person, is created when a sperm cell fertilizes an egg cell, the female gamete. The embryo sac's micropylar end, which serves as the entryway for the pollen tube to enter during pollination, is not far from the egg apparatus.

This arrangement facilitates the fertilization process by positioning the egg cell and synergids in close proximity to the site of pollen tube entry, increasing the chances of successful fertilization.

28. Correct option: (B) a – ii; b – i; c – iii; d – iv

a) Eyes of - ii) Potato:

The little depressions or buds that may be seen on a potato's surface are referred to as "eyes" in this context. These "eyes" are actually latent buds that, given the right conditions, can sprout and develop into new potato plants. Potatoes are a good example of vegetative multiplication through these bud structures because each eye has the capacity to produce a new plant.

b) Rhizome of - i) Ginger:

A rhizome is a form of modified stem that develops horizontally underground, and ginger is a rhizome. It functions as the plant's storage organ and is capable of producing new shoots and roots. A piece of ginger rhizome can sprout new shoots and grow into a new ginger plant when it is planted.

c) Bulbil of - iii) Agave:

On the plant's flower stalks, agave plants generate bulbils, which are tiny aerial structures. When these bulbils fall to the ground and take root, they might develop into new plants. Plants can create progeny without the use of seeds thanks to asexual reproduction known as bulbils.

d) Leaf buds of - iv) Bryophyllum:

Succulent bryophyllum reproduces vegetatively by means of its leaf buds. Small plantlets or buds develop around the borders of its leaves. These buds have the ability to root and develop into new plants when they fall to the ground or come into contact with soil. Bryophyllum and related plants are particularly prone to this type of proliferation.

29. Correct option: (C) a – iii; b – iv; c – i; d – ii

a) Fission - iii) Algae, fungi, Monera:

Asexual reproduction via fission occurs when an organism divides into two or more parts, each of which develops into a new individual. Algae, fungus, and Monera (bacteria) are examples of single-celled organisms that frequently go through this process, in which the parent cell divides to produce genetically identical daughter cells.

b) Budding - ii) Oedogonium, Chlamydomonas:

Asexual reproduction known as budding occurs when a new organism arises as a bud from the parent organism. This process is used by several algae, including Oedogonium and Chlamydomonas, where a tiny bud grows and ultimately separates to become a separate individual.

c) Fragmentation - i) Ulothrix, saprolegnia:

The splitting of an organism's body into multiple parts, each of which has the potential to develop into a new organism, is known as fragmentation. Water moulds like Saprolegnia and algae like Ulothrix can reproduce via fragmenting, in which parts of the parent plant separate and grow into new individuals.

d) Sporulation - iv) Diktyoma, fucus, Monera:

Sporulation is the process by which specialized reproductive cells with the capacity to develop into new organisms, known as spores, are formed. This process is seen in a variety of organisms, including certain bacteria (Monera) and algae (Dickyota and Fucus), where spores are created and spread to create new individuals.

30. Correct option: (B) (I), (iv)

Asexual reproduction techniques like stem cutting and tissue culture both produce offspring that are genetically related to the parent plant. In stem cutting, a section of the stem of a plant is removed, planted, and allowed to grow roots and become a new plant. Since the new plant is effectively a clone of the parent plant, this approach produces children that are genetically identical to the parent plant. Similar to this, tissue culture entails cultivating a little portion of plant tissue—such as a cell or a cluster of cells—in a sterile environment with nutrients and hormones in order to transform it into a new plant. Due to the fact that the new plants are created from the same original tissue sample, tissue culture also produces plants that are genetically identical.

As seeds are created through sexual reproduction including the union of male and female gametes, and mutations can result in alterations to a plant's genetic make-up, on the other hand, seed production (ii) and mutation (iii) bring genetic variety.

7. Reproduction in Animals

Solutions

LEVEL-I

1. Correct option: (C) Leopard

Animals that produce living offspring after the embryo has internally grown within the mother's body are said to be viviparous. The leopard in this instance is a viviparous animal. In the course of gestation, the foetus grows inside the mother's uterus in female leopards. The mother gives birth to live cubs once the foetus is fully formed. When compared to species that lay eggs or have less developed young, species like leopards benefit from this reproductive strategy because their young require a higher amount of protection and nurturing.

2. Correct option: (D) Cat

Mammals, including domestic cats, have epidermal hair, commonly referred to as fur or coat, which is a distinguishing characteristic. Cats have a thick layer of epidermal hair on their bodies, which the animals use for communication, camouflage, insulation, and other purposes. By trapping air near the skin, the hair aids cats in controlling their body temperature. In their native environments, it helps them blend in with their surroundings so they can stalk prey or avoid predators. They also receive sensory information about their surroundings through the tactile nature of cat fur, which improves their ability to navigate and interact with their environment.

3. Correct option: (D) All the above

Numerous creatures, including Hydra, algae, and fungi, have been found to reproduce asexually, which is a manner of reproduction devoid of gamete fusion. Simple aquatic organisms like the hydra reproduce asexually by budding. Photosynthetic organisms known as algae frequently reproduce asexually through fragmentation or spore production. Molds and yeasts are examples of fungi, which also use spore generation, budding, and fragmentation as asexual reproduction strategies. These species use asexual reproduction to rapidly produce young that are genetically identical or similar, allowing them to quickly adapt to their environment, fill new habitats, and continue to exist in different ecosystems.

4. Correct option: (B) Amoeba

Asexual reproduction known as binary fission occurs when a single parent cell divides into two identical daughter cells. A single-celled organism called an amoeba reproduces by binary fission. The amoeba's cytoplasm and nucleus divide during this phase, creating two daughter cells that are genetically identical to one another. Some unicellular creatures, like amoeba and bacteria, frequently reproduce in this way in order to proliferate and colonise their surroundings.

5. Correct option: (D) Stamen

The stamen, a flower's male reproductive organ, is absent from female flowers. The filament, which supports the anther, and the anther, which produces pollen grains, are the two primary components of the stamen. The pistil, which has three main components—the ovary, the style, and the stigma—is the reproductive structure in female flowers. Ovules are found in the ovary, where they undergo fertilisation to become seeds. The stigma is the receptive surface where pollen settles and germinates, leading to fertilisation, and the style is a tube-like structure that connects the stigma to the ovary.

6. Correct option: (B) Testes

The testes are one of the most important parts of the male reproductive system, which also comprises other organs. Sperm and male sex hormones like testosterone are produced by the testes. The epididymis, which matures and stores sperm, the vas deferens (sperm ducts),

which transport sperm from the testes to the urethra, and accessory glands that generate fluids that combine with sperm to make semen are further parts of the male reproductive system. The testes are unique to the male reproductive anatomy, in contrast to the other alternatives (Fallopian tube, uterus, and ovary), which are components of the female reproductive system.

7. Correct option: (C) Uterus

The uterus, a crucial reproductive organ in all female mammals, including humans, is where the embryo grows. The zygote (fertilised egg) passes through numerous cell divisions after fertilisation to become a blastocyst. The blastocyst is subsequently implanted into the uterine lining, where it receives nourishment and assistance for growth. During pregnancy, the uterus offers a favourable environment for the embryo to transform into a foetus. This process results in the development of diverse embryonic structures as well as the growth of organs and systems, which ultimately results in the birth of a child that is fully developed.

8. Correct option: (A) Frog

Fertilization normally occurs externally in frogs. Male and female frogs both discharge sperm and eggs into the water during the breeding season. This external fertilisation takes place in watery habitats like ponds, lakes, or streams when the female frog's eggs and the male frog's sperm come into touch. Tadpoles are produced from the fertilised eggs, and these creatures eventually change into adult frogs. This process of fertilisation, which is an illustration of external fertilisation as opposed to internal fertilisation, in which fertilisation takes place inside the female's body, is ideally adapted to the frog's aquatic existence.

9. Correct option: (B) Tadpole

Tadpoles are the name for a frog's larval stage. Aquatic animals known as tadpoles are the offspring of frog eggs. As they grow into adult frogs, they often have a long tail, external gills for breathing underwater, and go through a transformation process. Tadpoles eventually lose their gills during metamorphosis, and their tails shorten and their legs begin to develop. They eventually change into froglets, which then grow into adult frogs that can survive on land. Tadpoles are an important stage in the life cycle of the frog, allowing them to colonise aquatic habitats before switching to a terrestrial way of life.

10. Correct option: (A) Male flower

The male reproductive organs of blooming plants are called pollen grains. They are created by the stamen and other male flower components. Anthers and filament make up the stamen. The pollen grain, which contains the male gametes, is created by the anther (sperm cells). When pollen grains are discharged from the anthers, they are frequently carried to female flowers by wind, insects, or other pollinators where they can fertilise the ovules and start the seed-forming process.

11. Correct option: (C) In Vitro Fertilisation

In vitro fertilisation is a technique used to create "test tube babies," commonly referred to as "IVF babies." An egg (ova) is fertilised by a sperm during IVF outside of the body, typically in a lab dish or culture medium. After fertilisation, the resultant embryos are given a few days to mature before being placed into the uterus in preparation for possible implantation and pregnancy. When natural conception is difficult or not possible, IVF is a reproductive technology used to treat infertility and help people or couples become pregnant.

12. Correct option: (C) 270–280 days

Human gestation, sometimes referred to as pregnancy, usually lasts between 270 and 280 days, or around 9 months. From the first day of the woman's last menstrual period until the baby is born, this period is measured. The precise duration of pregnancy can vary slightly from woman to woman and can be impacted by a number of variables, including

the mother's and the baby's health, genetics, and environmental influences. However, the typical length is between 270 and 280 days.

13. Correct option: (D) Camel

The other choices, fish, frog, and lizard, are all vertebrate animals that fall within the classifications of Reptilia (for lizards) and Amphibia (for frogs) (for lizards). The physiological and anatomical properties of these animals are comparable. The camel, on the other hand, is a mammal and a member of the Mammalia phylum. All of the animals on the list are vertebrates, but the camel stands out from the other three due to its warm-bloodedness, the fact that it has mammary glands for producing milk, and its adaptations to desert settings. The camel differs from the other animals in that it is not aquatic, amphibious, or reptilian in nature.

14. Correct option: (D) Ostrich

The ostrich is an example of an oviparous animal, which are those that deposit eggs. African ostriches are enormous, flightless birds. They lay their eggs in ground nests, which are either incubated by the female ostrich or, in certain situations, also by the male. Prior to being laid, the eggs are internally fertilised, and they normally hatch after a predetermined incubation period. In order to secure the survival of their young in a variety of settings and conditions, oviparous animals, like the ostrich, have evolved this reproductive strategy. This technique frequently entails protective nests and systems to preserve the ideal environment for embryo development.

15. Correct option: (D) Reproduction

The fundamental life process that is essential to the survival of a species is reproduction. Through the development of children, it entails the creation of new individuals. In order to sustain genetic diversity and the survival of the species, reproduction makes sure that genetic information is transferred from one generation to the next. There are many different ways for different species to reproduce, including asexual reproduction, in which children are created from a single parent, and sexual reproduction, in which male and female gametes are fused to create offspring. A species' ability to reproduce and maintain its population over time would be compromised without it, which would eventually cause it to decrease and go extinct.

16. Correct option: (B) Ian Wilmut

An important historical milestone was reached when the first instance of cloning was carried out by pioneering British embryologist Ian Wilmut. He was an important contributor to the group at the Roslin Institute in Scotland that cloned a sheep dubbed "Dolly" successfully in 1996. Dolly was a significant accomplishment since she was the first mammal to be cloned utilising the somatic cell nuclear transfer method from an adult somatic cell. In addition to showing that it was possible to clone animals, Wilmut's discovery also provided new opportunities for genetic study, biotechnology, and the knowledge of cellular reprogramming. The study of biology and genetics advanced significantly as a result of this ground-breaking achievement.

17. Correct option: (C) Metamorphosis

A larval stage of an organism goes through considerable structural and physiological changes during the process of metamorphosis to become an adult form. This metamorphosis involves a number of developmental phases, many of which have unique external characteristics and ecological functions. The majority of insects, amphibians, and some aquatic species exhibit this behaviour. Examples of metamorphosis include the development of a tadpole into a frog or a caterpillar into a butterfly. In order to adjust to the new adult lifestyle and surroundings, it often entails the reorganisation and reabsorption of tissues as well as the restructuring of bodily parts.

18. Correct option: (A) Asexual

Asexual reproduction is defined as reproduction that occurs without the release of male and female gametes. Asexual reproduction produces children without the fusing of gametes, which are specialised reproductive cells, from two distinct parents. Since there is no genetic material mixing as occurs during sexual reproduction, this technique produces children that are genetically identical or highly similar. Binary fission, budding, and fragmentation are asexual reproduction techniques examples where the offspring receive the genetic features straight from a single parent.

19. Correct option: (D) Hermaphrodite and Bisexual animal

Hermaphrodites are animals with both male and female reproductive systems. As a result, they can reproduce with either males or females of their species since they have the capacity to generate both eggs and sperm. When referring to an animal that can reproduce in both sexes, the terms "bisexual animal" and "hermaphrodite" are frequently used interchangeably. As a result, D) Hermaphrodite and Bisexual Animal is the right response to the question because it includes both names and depicts creatures with both male and female reproductive systems.

20. Correct option: (B) Budding

In yeast cells, asexual reproduction occurs by budding. A tiny bud or protrusion on the parent cell emerges during budding, and as it expands and separates from the parent cell, it develops into a new organism. Under ideal circumstances, this mechanism enables yeast cells to multiply quickly and successfully. Yeast cells benefit from budding because it enables them to rapidly increase their population and adapt to changing conditions. Additionally, because the offspring receives a copy of the parent's genetic material, budding ensures genetic resemblance between the parent and offspring. Overall, yeast cells primarily reproduce asexually by budding, which aids in their survival and growth.

LEVEL – II

1. Correct option: (B) Reproduction

The biological process of reproduction is how living things create new members of the same species. It is a basic aspect of life that guarantees the survival of species over generations. It is possible for creatures to reproduce asexually and sexually, and this allows them to pass on their genetic makeup and characteristics to the following generation. The ability to make new people of the same sort is not immediately related to respiration, on the other hand, which is a metabolic activity involved in energy production and the exchange of gases.

2. Correct option: (A) 2

Asexual reproduction and sexual reproduction are the two basic types of reproduction. Asexual reproduction refers to the generation of genetically identical or extremely similar children from a single parent organism. Contrarily, sexual reproduction entails the union of gametes (sperm and eggs) from two distinct parents, resulting in genetic variation in the progeny. These two methods of reproduction are important tactics used by different creatures to maintain their species.

3. Correct option: (A) Asexual

One parent usually participates in asexual reproduction, which is a sort of reproduction. Asexual reproduction is the process of producing offspring from a single parent organism without the joining of gametes or the use of genetic material from another person. Since there is no recombination of genetic material as occurs during sexual reproduction, this technique produces children who are genetically identical or highly similar. Numerous

organisms have asexual reproduction, which is advantageous in circumstances that favour rapid population increase and colonisation.

4. Correct option: (B) Asexual reproduction

Amoebas are single-celled organisms that frequently reproduce asexually through a process known as binary fission. In binary fission, the amoeba's cell nucleus splits in half, then the remaining components of the cell divide as well. Due to this, two daughter cells are created that are genetically identical to one another and grow and develop into separate amoebas. Under the right circumstances, asexual reproduction enables amoebas to multiply quickly and colonise their surroundings.

5. Correct option: (C) Budding

The Cnidaria phylum member Hydra is a basic aquatic organism that reproduces asexually via a process known as budding. A new person, known as a bud, develops as a protrusion on the parent organism during this process. The bud gradually matures and separates from the parent to become a distinct and autonomous organism. Hydra frequently reproduce asexually by budding, which enables them to effectively increase their number under favourable circumstances.

6. Correct option: (D) Hydra

When an organism reproduces asexually, it creates offspring that are genetically identical to it. Simple creatures like bacteria and plants frequently reproduce in this way. Asexual reproduction is possible for the small aquatic organism Hydra, which is a member of the phylum Cnidaria. This process is known as budding. A new individual develops as an outgrowth or bud on the parent organism during budding. This bud develops, eventually separating to become a separate entity. Asexual reproduction through budding allows hydra to efficiently produce genetically identical offspring, promoting population growth in suitable environments.

7. Correct option: (B) Viviparous

The animals that directly reproduce young ones are called viviparous. Viviparity is a form of reproduction where the embryos grow inside the mother's body and are fed by the mother herself. The growing embryos in viviparous creatures, such as most mammals and certain reptiles, are given nutrition and oxygen through a placental link or other specialised structures, allowing them to grow and develop before giving birth as living offspring. Opposing this mode of reproduction is oviparity, in which animals lay their eggs outside of their bodies, and the embryos grow outside the mother's body. Therefore, the correct answer is B) Viviparous.

8. Correct option: (D) Sperm

Sperm makes up the male gamete. In many animal species, including humans, males generate sperm, which are specialised reproductive cells. Sperm cells are created with fertilisation in mind. They often have a tail (flagellum), which allows them to travel through fluid environments like the female reproductive canal to reach the egg. Sperm are essential for sexual reproduction because their nuclei contain genetic information. The formation of a new person with a special blend of genetic features from both parents begins with the union of a sperm and a female gamete (egg), which creates a zygote. Therefore, sperm are necessary for a species to continue through sexual reproduction.

9. Correct option: (A) Sponges

Internal budding, commonly referred to as gemmulation, is an asexual reproduction process that is seen in sponges (phylum Porifera). Gemmules, which are tiny bud-like structures, develop within the parent sponge's body throughout this phase. The cells that make up these gemmules are grouped together and covered in a layer of defence. The parent sponge can discharge these gemmules into the nearby water when the environment becomes

adverse, such as during periods of hard weather. The gemmules serve as a kind of survival capsule, shielding the sponge cells from harmful elements. When conditions are better, the gemmules can separate and grow into new sponge individuals, enabling sponges to endure and spread over a variety of aquatic settings. Internal budding in sponges is a remarkable adaptation that aids in their ability to reproduce asexually and withstand challenging ecological circumstances.

10. Correct option: (C) Fragmentation

Asexual reproduction known as fragmentation allows for the growth of a whole new organism from specific parts or cells of the original organism. The parent organism splits into various pieces throughout this process, and each piece has the ability to regenerate and grow into a brand-new, autonomous individual. Each fragment finally develops into a complete organism after the missing body parts are grown again by the pieces. Many different animals, including flatworms and some plant species, exhibit fragmentation, which enables them to procreate without the necessity for mating or the development of specific reproductive cells. The parent organism can produce several offspring using its own body parts thanks to this reproductive strategy, which increases population size.

11. Correct option: (B) Fragmentation.

Flatworms, also known as planarians, are capable of reproducing through a process called fragmentation. A flatworm can be divided into two or more parts via this technique of reproduction, and each portion can grow back into a whole creature. Each piece develops into a distinct individual and grows any missing body parts, such as the head and tail. Flatworms can produce new individuals from dismembered or separated body parts thanks to this type of asexual reproduction. While the animal kingdom also employs alternative asexual reproduction strategies, such as binary fission and budding, flatworms stand out for their use of fragmentation.

12. Correct option: (B) Sexual reproduction

An example of sexual reproduction occurs when male and female gametes from two distinct parents are fused. Genetic material is transferred between the parents during sexual reproduction, resulting in a genetically diverse progeny. This procedure typically calls for the development of specialised reproductive cells, or gametes, like sperm and eggs, which have half as many chromosomes as usual. These gametes join together during fertilisation to form a new person that combines genetic characteristics from both parents. Contrarily, asexual reproduction entails the generation of children without the requirement for gamete fusion, resulting in offspring that are genetically identical to their parents. B) Sexual reproduction is the right response, so.

13. Correct option: (C) Zygote

A sperm or egg (ova) cell is an example of a specialised reproductive cell known as a gamete. Gametes are necessary for sexual reproduction and are produced by both male and female members of a species. A zygote is the structure formed when a male gamete (sperm) combines with a female gamete (egg). The first cell to form during fertilisation is called a zygote, and it combines genetic material from both parents. The embryo develops into a multicellular organism after undergoing fast cell divisions and development from the single-celled zygote. Thus, the development of sexual reproduction begins with the union of two gametes to produce a zygote.

14. Correct option: (B) Bisexual organism

An organism that has both male and female reproductive systems and is able to produce both male and female gametes (sperm and eggs) within a single individual is referred to as a hermaphrodite or a bisexual organism. With the use of one's own sperm and eggs, an individual may self-fertilize via this reproductive technique. Many animal species,

particularly invertebrates like some fish, worms, and snails, are found to be hermaphrodites. The ability to reproduce even without a mate is ensured by this technique, however genetic variety may be lessened compared to cross-fertilization between different individuals.
Unisexual organisms, on the other hand, produce only one type of gamete. Male organisms produce sperm, and female organisms produce eggs. Some examples of unisexual organisms include humans, dogs, and cats.

15. Correct option: (C) Testes

As a component of the male reproductive system, the testes are where sperm, also known as spermatozoa, are created. The testes are two oval-shaped organs that hang outside of the male body in the scrotum. Seminiferous tubules, a large number of small tubules within the testes, are where spermatogenesis occurs. Sperms are created during the process known as spermatogenesis from unique cells known as spermatogonia. These cells go through a number of divisions and changes before becoming fully developed sperm cells. The mature sperm are released into the epididymis, a coiled tube attached to the testes, where they continue to develop and are kept until ejaculation, once they have fully developed. In the male reproductive system, the testes serve as the major location for the generation of sperm.

16. Correct option: (C) Sperm ducts

Sperm are transported from the testes to the penis by sperm ducts, commonly known as the vas deferens or ductus deferens. The male reproductive system includes these ducts, which act as a route for sperm to follow after ejaculation. The testes produce sperm, which mature as they pass through the testes' epididymis, a coil-like structure. In order to get to the urethra near the base of the penis, the sperm must first travel through the sperm ducts, which run from the epididymis through the pelvic cavity. The sperm are launched through the sperm ducts during ejaculation and combine with secretions from other accessory glands to form semen, which is then ejected from the penis. Thus, the sperm ducts play a crucial role in the transport of sperm for fertilization.

17. Correct option: (B) Four

A specialised cell used in sexual reproduction is a sperm. The head, the midpiece, the tail (sometimes referred to as the flagellum), and the neck are its four primary components. The genetic material is housed in the compact nucleus of the cranium. During fertilisation, the acrosome, a structure at the tip of the head, provides enzymes required for the sperm to pierce the egg's outer layers. There are many mitochondria in the midpiece, which supply the sperm with the energy it needs to migrate. The sperm can move through fluid environments thanks to the tail, or flagellum, which is a long, whip-like structure. The midpiece and the head are connected by the neck. These four components work together to facilitate the sperm's journey toward the egg for the potential process of fertilization.

18. Correct option: (B) Female

The correct answer is B) Female. The flagellum, or tail, of a sperm is a whip-like feature that helps the sperm move forward in a fluid environment. The sperm's tail is principally in charge of enabling travel into the female reproductive canal in the context of human reproduction. The sperm must pass through the female reproductive system after being ejaculated during sexual activity in order to get to the fallopian tubes, where fertilisation may eventually take place. The sperm can travel via the cervical mucus, the uterus, and finally the fallopian tube where they may come into contact with an egg for fertilisation thanks to the movement of the tail. As a result, the tail's main job is to facilitate sperm passage inside the female reproductive system.

19. Correct option: (B) Penetration

A sperm cell's head contains a unique component called the acrosome. It is essential to the fertilisation process. When the sperm makes contact with the zona pellucida, the outer covering of the egg, enzymes from the acrosome, which are present, are released. These enzymes aid in removing the egg's defensive barriers, allowing the sperm to enter and unite with the egg cell. This penetration is a crucial stage in fertilisation because it enables the sperm's genetic material to unite with the egg's genetic material to create a zygote, which will later develop into an embryo. As a result, the acrosome's main purpose is to make it easier for the egg to penetrate.

20. Correct option: (A) Ova

The female reproductive system's egg-producing organs are the ovaries. A different name for the eggs is ova. One ovary is situated on either side of the uterus in the pelvic. The female gametes or eggs, known as ova, are produced by the ovaries, which are a component of the female reproductive system. These eggs, which contain the genetic material of the female, are necessary for sexual reproduction. Each of the countless ovarian follicles found in ovaries is home to an immature egg. Typically, one egg is chosen and matures within a follicle during the menstrual cycle before being released through the process of ovulation. Following the release of the egg and the subsequent fusing of a sperm and an egg, fertilisation can result in the creation of a foetus.

21. Correct option: (B) Ovulation

A developed egg, or ovum, is released from the ovary into the fallopian tube during ovulation, a biological process. The pelvis contains two tiny, almond-shaped structures called ovaries. They generate hormones and eggs. The ovarian follicle that contains the egg ruptures during ovulation, releasing the egg into the pelvic cavity. The fallopian tube, which is lined with small hair-like structures known as cilia and aids in capturing the egg there, then directs it toward the uterus. The sperm cell must fertilise the egg for it to embed in the uterine lining and begin a pregnancy. Ovulation often occurs 14 days or so before a woman's period begins.

22. Correct option: (C) Inside the female body

Male gametes (sperm) are placed inside the female body during internal fertilisation, where they fertilise the female gametes (eggs). Mammals, birds, and some reptiles frequently undergo this sort of fertilisation. In mammals, the male ejaculates sperm after inserting his penis into the female's vagina. The sperm next proceed to the ovaries, where they can fertilise the eggs, via the uterus and fallopian tubes. Compared to external fertilisation, internal fertilisation provides a number of benefits. The fact that the eggs are shielded from predators and the elements is one benefit. The sperm are more likely to fertilise the eggs, which is an additional benefit.

23. Correct option: (A) Fertilization

The creation of a zygote occurs when a male gamete (sperm) combines with a female gamete (egg or ovum) during the process of fertilisation. This is the first step in a person's growth. A sperm cell enters the interior of the egg during fertilisation after breaking through its outer layers of protection. Once within, the sperm and egg nuclei combine to form a single cell, fusing their genetic material. The two parents' genetic material combines to form a unique set of DNA that determines the offspring's genetic features and characteristics. The crucial process of fertilisation takes place inside the female reproductive system, frequently in the fallopian tubes in humans.

24. Correct option: (B) Fallopian tube

Human fallopian tubes are where fertilisation occurs. The slender, tube-like fallopian tube, often referred to as the oviduct, joins the ovaries with the uterus. An egg is discharged from

one of the ovaries during ovulation, and it travels through the fallopian tube. Fertilization can take place if both sperm are in the fallopian tube at the same moment. To get to the fallopian tube, the sperm must pass through the cervix and enter the uterus. The fertilised egg, also known as a zygote, travels via the fallopian tube after fertilisation in order to reach the uterus, where it implants and grows into a foetus. Therefore, human fertilisation takes place in the fallopian tube.

25. Correct option: (A) Fish

In fish, external fertilisation is seen. The male fish releases his sperm to externally fertilise the eggs while the female fish discharges her eggs into the water throughout this procedure. This sort of fertilisation takes place outside the female fish's body, typically in bodies of water like rivers, lakes, or oceans. Fish benefit from external fertilisation because it makes it possible to fertilise many eggs at once, increasing the likelihood of fertile offspring. Additionally, it gives sperm and eggs a chance to freely mix in the water, enhancing the genetic diversity of the offspring. However, external fertilisation also subjects the sperm and eggs to a variety of environmental elements, leaving them open to predators and other dangers. Overall, external fertilisation is a unique reproductive strategy observed specifically in fish

26. Correct option: (C) Implantation

The procedure through which the embryo fuses to the uterine wall is known as implantation. It is an important stage in the progression of a pregnancy. The endometrium, the uterus' lining, is where the embryo first penetrates. The embryo's trophoblast cells then infiltrate the endometrium and establish a blood vessel between it and the mother. This process, which enables the development of the placenta, a crucial organ created later in gestation, allows the embryo to obtain needed nutrients and oxygen from the mother's bloodstream, is crucial for the progression of pregnancy. In most cases, implanting happens 6 to 10 days after conception. After fertilisation, it can happen as soon as 4 days later or as late as 12 days later.

27. Correct option: (B) Foetus

The term "Fetus" refers to the developmental stage at which an embryo's individual bodily components may be recognised. The embryo has grown and differentiated significantly during this stage, and its main organs and structures have formed. Rapid cell division and the building of the framework for many organs and systems characterise the development in the stage before this one, known as the embryo stage. The embryo's fundamental structures have formed during the foetal stage, which is a later stage of development.

28. Correct option: (B) Uterus

The uterus is not a part of the male reproductive organ. It is a female reproductive system that is essential to pregnancy. During pregnancy, the uterus serves as the foetus' home and source of nutrition. The reproductive system in males consists of the penis, testes, and seminal vesicles. Urine and semen are transported through the penis during sexual activity. The seminal vesicles help to create semen, whereas the testes produce sperm and male sex hormones. The only organ available that is not a part of the male reproductive system is the uterus.

29. Correct option: (D) Both B and C

Animals that give birth to live offspring are said to be viviparous. Animals that lay eggs include hens and silkworms, which are both oviparous. Young that can survive without their mother emerge from the eggs.

Eggs laid by hens develop into chicks. Within hours of hatching, the chicks are independent. Eggs laid by silkworms develop into caterpillars. The caterpillars undergo several moults before pupating to become moths. The moths have the ability to fly and lay eggs.

Humans are viviparous creatures. The fertilised egg grows into a foetus after implanting in the mother's uterus. Through the placenta, the mother feeds the foetus. After roughly nine months of gestation, the baby is delivered.

30. Correct option: (D) Hen

Hens do not undergo metamorphosis, a process that results in a significant shift in body composition. Hens grow and mature gradually without going through the specific metamorphic changes that these other organisms do, but animals like silkworms, fish, and butterflies go through metamorphic transformations during their life cycles.

Silkworms and butterflies emerge from their eggs as caterpillars, and as they develop, they undergo a series of moults. While the butterfly's last moult results in the adult butterfly, which has wings, the silkworm's final moult results in the adult silkworm, which is a moth. Fish grow through a series of moults after emerging from their eggs as larvae. The adult fish, which has scales and fins, is created after the last moult.

8. Reaching the Age of Adolescence

Solutions

LEVEL -I

1. Correct option: (B) 10–19 years

Between childhood and adulthood, adolescence is a period of profound physical, cognitive, and emotional change. It normally starts at age 10 and lasts until around age 18 or 19. People go through puberty during this time, which involves the development of secondary sexual traits and the capacity for reproduction. Adolescents also experience considerable brain development, which enhances their capacity for logic, judgement, and self-control. On an emotional level, adolescents might go through mood changes, increased self-consciousness, and the development of a personal identity. Given these considerations, the correct response to the question is B), which covers the age span in which the majority of teenagers go through these transformative changes.

2. Correct option: (C) Peers

Adolescents are at a developmental stage where they are seeking independence and creating their own identities, so it makes sense that they would be interested in spending time with their peers. Peers give teenagers a sense of community and acceptance, enabling them to explore their shared interests, morals, and worldviews. In addition to providing opportunities for social and emotional development, peer interactions help young people develop critical abilities including empathy, communication, and dispute resolution. Adolescents can share experiences and activities with their peers, which promotes a sense of togetherness and support for one another. In general, teenage social and emotional development depends heavily on peer relationships.

3. Correct option: (A) Larynx

Particularly in men, the Adam's apple is a noticeable feature of the neck. The thyroid cartilage, a component of the voice box or larynx, forms it. The larynx, which is found in the throat, is important for both creating sound and allowing air to reach the lungs. Due to the effects of testosterone throughout puberty, the laryngeal prominence, often known as the Adam's apple, is more pronounced in males. It helps to produce deep voices and acts as a protective structure for the vocal cords. Larynx is the proper response, so that is what it is. The alternate choices are false. The mouth is the aperture at the front, the pharynx is the throat, and the oesophagus is the food pipe.

4. Correct option: (B) Gametes

The reproductive organs create gametes, which are reproductive cells. The testes are the male reproductive system. The male gametes, or sperm, are created in the testes. The ovaries are a female's reproductive organs. The female gametes, or eggs, are created in the ovaries. A gamete is required for reproduction. A zygote, which is an egg that has been fertilized, is created when a sperm and an egg join. The zygote can then grow into a brand-new person.
The alternate choices are false. Chemicals known as hormones are created by the endocrine system. The male reproductive cells of plants are called pollen grains.

5. Correct option: (D) 21 years

The legal age for female marriage is 21. This means that a girl must be at least 21 years old in order to lawfully get married. In order to ensure that people are capable of making such an important decision and completely comprehend the obligations and repercussions that come with marriage, the legislation has imposed this age restriction. It also aims to safeguard young women from being pressured into marriages at an early

age, giving them more time to pursue their education, job goals, and personal development before committing to a committed life partner.

6. Correct option: (C) Endocrine gland

The answer to the question is C) Endocrine gland. Hormones are produced and released into the bloodstream by endocrine glands. These hormones are essential for the growth and control of the human reproductive system. They manage the development and maturation of reproductive organs such the ovaries and testes, as well as the gamete production (sperm and eggs). Endocrine glands secrete hormones like oestrogen and testosterone, which are crucial for the maturation of secondary sexual traits, the onset of puberty, and the maintenance of reproductive capabilities into adulthood. Endocrine glands are therefore essential for the growth and operation of the human reproductive system.

7. Correct option: (D) All of the above

Digestion, respiration, and growth are a few of the different processes that make up an organism's character. The term "digestion" describes the process by which food is broken down into smaller molecules that the body can absorb and use. Gases like oxygen and carbon dioxide are exchanged during respiration to create energy. The term "growth" describes how an organism's size or cell count increases. The existence and operation of living things depends on each of these processes. Since all of these processes contribute to the general nature of living things, the correct response is D) All of the above.

8. Correct option: (C) Pituitary

Emotional regulation is the responsibility of the pituitary gland. Due to its ability to create and release hormones that control a variety of biological processes, including emotions, it is frequently referred to as the "master gland." At the base of the brain, the pituitary gland connects to the hypothalamus. It releases hormones that have an impact on stress levels, mood, and general mental health. C) Pituitary is the right response to the question.

The alternate choices are false. The male gonads are the testes, while the female gonads are the ovaries. Although they create sex hormones, they do not have emotional control. A hormone called adrenaline is released in response to stress, yet it has little effect on mood.

9. Correct option: (C) Adolescence

The period between childhood and adulthood is known as adolescence. People go through major physical, emotional, and cognitive changes throughout this transitory stage. Teenagers go through puberty during this time, which entails the emergence of secondary sexual traits. Additionally, they go through heightened emotions, greater independence, and an identity crisis. As people start to create their own opinions, values, and social networks, adolescence is a key time for social and emotional growth. It's a period of exploration, self-discovery, and getting ready for the obligations and difficulties of maturity. Adolescence is the proper response, which is C.

10. Correct option: (B) Maximum

Children go through a considerable growth spurt during adolescence, which results in an increase in height. Numerous elements, including nutrition, hormones, and heredity, have an impact on this growth spurt. It's common to refer to a child's height during this time as their maximum height. This is due to the fact that growth slows and eventually stops after adolescence, resulting in the establishment of their adult height. As a result, B) Maximum is the correct response to the question because it captures the maximum height that children can reach during adolescence.

11. Correct option: (A) 25%

The percentage of smokers in India is A) 25%, in response to the question. This indicates that in India, there are about 25 percent of smokers. The fact that such a high percentage of the population smokes, suggesting that smoking is a widespread problem in the nation,

should not be overlooked. The high rate of smoking emphasizes the need for strong policies to discourage smoking, advance public health, and run awareness efforts to lessen the negative impacts of smoking on people and society at large.

12. Correct option: (A) Adam's apple

The Adam's apple, a bulge of cartilage on the front of the neck, is more noticeable in men than in women. It is brought on by the thyroid cartilage, one of the cartilages that make up the larynx, growing. Another name for the larynx is the voice box. Sound is produced by it, which is situated in the neck. To make sound, the two tissue bands that span the larynx called the vocal cords vibrate. There is no sound coming from the Adam's apple. But when the thyroid cartilage grows, it may enlarge and become more noticeable, giving the voice a deeper tone.

The other options are incorrect. The voice box is the complete larynx, and the voice chamber is a different name for the larynx. Thyroid is a gland that is located in the neck, but it is not part of the larynx.

13. Correct option: (B) Sebaceous glands

Sebaceous glands are another name for oil glands. These glands are in charge of creating sebum, an oily material that is vital for lubricating the skin and hair. Sebum aids in retaining moisture in the skin and preventing drying out. Additionally, it aids in defending the skin from aggressors including bacteria and toxins from the environment. Except for the palms of the hands and the soles of the feet, the body is covered in sebaceous glands. On the face, scalp, and upper back, they are most prevalent. Oily skin and acne can result from these glands being hyperactive.

The alternate choices are false. Lacrimal glands create tears, sweat glands create perspiration, and serumenous glands create earwax.

14. Correct option: (C) Menarche

The right response is C) Menarche. The term "menarche" refers to a girl's first period, which normally begins when she is between the ages of 12 and 14. It signals the start of female puberty and the start of the reproductive stage. Other puberty symptoms, like breast development, pubic hair growth, and broadening of the hips, are frequently present when a woman experiences her first menstrual period. There may be some discomfort during menarche, such as cramping and bloating.

Contrarily, menopause is the term used to describe the end of women's menstruation and fertility, which typically happens between the ages of 45 and 55. A freshwater ecosystem's succession of plant species is referred to as a "hydrarch" in ecology.

15. Correct option: (C) 21

For men, 21 is the minimum age to get married. This means that in order for a man to lawfully engage into a marriage contract, he must be at least 21 years old. In order to ensure that people have attained a particular degree of maturity and are capable of making thoughtful decisions on their marriage, the law has imposed this age requirement. Men are said to be emotionally and mentally ready for the obligations and commitments that come with marriage by the age of 21, having adequate life experience. This age restriction also guards against the possibility of exploiting or pressuring younger people into marriages before they are ready.

16. Correct option: (C) Iron

Because iron is needed for the creation of hemoglobin, the protein that carries oxygen to the body's numerous tissues and organs, red blood cells (RBCs) are produced as a result. Iron deficiency anemia is a disorder that develops when the body is unable to create enough hemoglobin due to a lack of iron in the body. Shortness of breath, weakness, and

exhaustion are some of the signs of this illness, which leads to decreased RBC production. Iron is therefore essential for both the creation of RBCs and the body's general ability to transport oxygen.
The alternate choices are false. It is not necessary for the creation of RBCs to have magnesium. Additionally, neither boron nor zinc are required for the creation of RBCs, but they do have a role in other physiological processes.

17. Correct option: (B) Adolescents

Secondary sexual characters are seen in adolescents. During adolescence, individuals undergo significant physical changes as they transition from childhood to adulthood. These changes are driven by the activation of sex hormones, such as testosterone and estrogen, which lead to the development of secondary sexual characteristics. These characteristics include the growth of facial and body hair in males, breast development in females, deepening of the voice in males, and widening of the hips in females. These changes are essential for reproductive purposes and help distinguish individuals as sexually mature adults.

LEVEL -II

1. Correct option: (C) 19

The period between childhood and adulthood known as adolescence is characterized by various physical, emotional, and social changes. It is generally accepted to start around the time of puberty, which typically occurs when a child is 11 years old, and last until they are in their late teens or early twenties. Although there may be some variance in the precise age range, 19 is typically regarded as the upper age limit of adolescence. People are often on the verge of adulthood at this age, with many already taking on adult duties and making significant life decisions.
Adolescence brings about a variety of physical, mental, and emotional changes that can be difficult for both the teenager and their parents to deal with. The role of parents during this time should be one of understanding and support. They can also help their adolescent by providing them with information about puberty, healthy relationships, and decision-making.

2. Correct option: (B) Hormones

Chemicals called hormones are injected right into the bloodstream. They are created by a number of bodily glands, including the pituitary, thyroid, and adrenal glands. As messengers, hormones transmit signals from one area of the body to another. They control a variety of vital bodily processes, such as metabolism, reproduction, mood, and growth and development. Hormones go to their intended tissues or organs after being released into the bloodstream, where they attach to particular receptors and start a reaction. Hormones may connect with various bodily parts rapidly and effectively because to this direct release into the bloodstream, maintaining correct coordination and functioning.
The other options are incorrect. Blood is not a chemical substance, and lymph fluid is not released directly into the bloodstream.

3. Correct option: (B) Hormones

Glands create hormones in the body. The body's glands are specialized organs that release hormones into the blood. These hormones serve as chemical messengers and are essential for controlling a number of body processes and functions. They aid in the regulation of many physiological processes, including growth, metabolism, reproduction,

mood, and many more. The body couldn't sustain homeostasis and function normally without hormones.

A substance that travels throughout the body is called blood. Red blood cells, white blood cells, platelets, and plasma make up its composition. A transparent fluid called lymph fluid circulates throughout the lymphatic system. It aids in infection prevention and waste product removal from the body.

4. Correct option: (A) Endocrine

Endocrine is the correct answer. Since endocrine glands lack ducts or tubes to carry their secretions, they are also referred to as ductless glands. Instead, they immediately release hormones into the circulation. These hormones are subsequently transported to the body's target cells or organs, where they control a variety of physiological functions and uphold homeostasis. The pituitary gland, thyroid, adrenal, and pancreatic glands are a few examples of endocrine glands. These glands are vital for controlling the body's vital processes as growth, metabolism, reproduction, and others. Contrarily, exocrine glands do contain ducts that direct their secretions to a particular place. Exocrine glands include, for instance, salivary and sweat glands.

5. Correct option: (D) Brain

The pituitary gland is located at the base of the brain. It is a tiny, pea-sized gland that is frequently called the "master gland" since it is so important in controlling several organ functions. The pituitary stalk, a stalk-like structure, connects the pituitary gland to the hypothalamus, a brain region. Hormones that regulate development, metabolism, reproduction, and other critical bodily functions are created and released by this gland. It can receive signals from the hypothalamus thanks to its position at the base of the brain, and it can react by releasing the appropriate hormones into the bloodstream.

6. Correct option: (A) pituitary gland

At the base of the brain, there lies a pea-sized gland called the pituitary. It generates hormones that regulate the activity of other glands, such as the thyroid gland, adrenal glands, and ovaries, which is why it is known as the "master gland." Growth hormone is one of the hormones generated by the pituitary gland (GH). GH is in charge of promoting the development of bones and muscles. It also affects the growth of other tissues, including the liver, lungs, and heart. Growth hormone is not secreted by the pancreas, testes, or any of the mentioned organs.

7. Correct option: (C) Goitre

Goitre is caused by a thyroxine deficit. The thyroid gland produces the hormone thyroxine, which is essential for controlling metabolism. Thyroxine shortage causes the thyroid gland to expand in an effort to make up for the reduced hormone output. Goitre is the medical term for this thyroid gland hypertrophy. Goitre can result in a neck swelling that is apparent and create symptoms including breathing or swallowing problems. C. Goitre is the appropriate response to the question.

8. Correct option: (A) Kidneys

The small, triangular-shaped adrenal glands are found on top of each kidney. They are a vital component of the endocrine system and produce and release hormones like cortisol, aldosterone, and adrenaline. The adrenal glands' location above the kidneys gives them direct access to the bloodstream, which facilitates the speedy and effective transportation of the hormones they manufacture throughout the body. Due to their near proximity, the kidneys and adrenal glands can communicate with one another and coordinate their efforts to control many biological processes like blood pressure, metabolism, and stress response.

9. Correct option: (C) Oestrogen and Progesterone

The female sex hormones, notably oestrogen and progesterone, are produced by the ovaries in females. Oestrogen is essential for the growth and control of the female reproductive system, as well as for the development of the breasts and the menstrual cycle. Contrarily, progesterone plays a role in getting the uterus ready for pregnancy and keeping a healthy pregnancy. Together, these hormones control ovulation, regulate the menstrual cycle, and aid in the growth and upkeep of the female reproductive system.

10. Correct option: (A) Testes

The male sex hormone testosterone is created by the testes, which are the male reproductive organs. These scrotal glands are essential for the growth of male reproductive characteristics and abilities. The expansion of the male reproductive organs and the development of secondary sexual features including face and body hair, a deeper voice, and these other sexual traits all depend on testosterone. Along with controlling sperm generation, muscle mass, bone density, and sexual health in general, it also controls a number of physiological functions.

Options B (Ovary) and C (Penis) are erroneous because they have nothing to do with testosterone production. Given that males' testicles do indeed create testosterone, Option D (None) is equally erroneous.

11. Correct option: (B) Sperm

Testosterone, the primary male sex hormone, plays a pivotal role in the development and maintenance of male reproductive traits and functions. It stimulates the production of sperm cells in the testes through a process called spermatogenesis. Sperm cells are crucial for male fertility and are responsible for carrying genetic information to fertilize an egg cell during sexual reproduction. While testosterone has other physiological effects in the body as well, its role in supporting the production of sperm cells is specifically significant in the context of male reproductive health.

Options A and C (Ova and Both A and B) are incorrect as they pertain to female reproductive components and do not align with testosterone's role in sperm production. Option D (None) is also incorrect given the established connection between testosterone and sperm cell development.

12. Correct option: (B) Pancreas

An essential organ in the human body that is necessary in controlling blood sugar levels is the pancreas. It secretes the hormones insulin and glucagon, which cooperate to regulate the metabolism of glucose. Glucagon boosts blood sugar levels by encouraging the release of glucose from the liver, whereas insulin helps lower blood sugar levels by enabling the uptake of glucose into cells. In order to regulate blood sugar levels and preserve overall health, there must be a dynamic interaction between the hormone’s insulin and glucagon.

The other options (A. Testes, C. Ova, D. Sperm) are not relevant to the secretion of insulin and glucagon.

13. Correct option: (C) Diabetes

The chronic medical illness known as diabetes is characterized by the body's inability to generate or use insulin, a hormone essential for controlling blood sugar levels. Type 1 diabetes and type 2 diabetes are the two most common kinds of the disease. In people with type 1 diabetes, the immune system of the body assaults and kills the cells that make insulin in the pancreas, which results in a lack of insulin. In type 2 diabetes, the body either stops producing enough insulin to meet the body's needs or develops a resistance to the effects of insulin. As a result, blood sugar levels increase, leading to a number of health issues. Maintaining blood sugar levels within a healthy range requires using a combination of food, exercise, medication, and, when necessary, insulin therapy.

14. Correct option: (A) Secondary sexual characters.

Between childhood and adulthood, adolescence is a period of profound physical, psychological, and emotional change. Secondary sexual traits start to show throughout this stage as a result of hormonal changes. These traits, which are particular to each gender and play a role in sexual maturation and differentiation rather than being directly related to reproduction, are gender-specific. Male secondary sexual characteristics can manifest as voice deepening, a rise in face and body hair, an increase in muscular mass, and alterations in body composition. These changes in females include the expansion of the breasts, broadening of the hips, and pubic and underarm hair growth. These traits set males and females apart, and they also contribute to attraction and reproduction, but they are different from primary sexual traits, which concern the reproductive organs directly.

15. Correct option: (D) All the above

Physical alterations brought on by hormonal changes during puberty are known as secondary sexual features. Males and females can be distinguished from one another based on these changes, which are suggestive of sexual development but are not directly related to the reproductive organs themselves. The voice alters as the larynx (voice box) develops, giving men a deeper voice. The growth spurt that comes with puberty, when bones lengthen, is what causes height increases. Both men and women can grow pubic hair, also known as genital hair, however the patterns and rates of growth might vary. These traits contribute to the total physical alteration, along with other modifications like the development of breasts in women and broader shoulders in men.

16. Correct option: (B) Menarche

Menarche, which normally occurs between the ages of 10 and 14, is the first time an adolescent girl has menstruation. The beginning of the female reproductive years is a crucial milestone. A girl's body experiences hormonal changes during menarche that cause the reproductive system to mature, including the development of the uterus and ovaries. When a girl's menstrual cycle starts, her body is ready to start releasing eggs and prepare for a possible pregnancy. Menarche varies from person to person and is affected by things like genetics, diet, and general health. It's a significant occasion in a young woman's life because it represents her entry into adulthood and the possibility of becoming pregnant.

17. Correct option: (A) Menopause

The menopause is a normal biological process that signals the end of a woman's reproductive years. It happens when a woman's ovaries stop producing eggs, which results in a permanent end to her monthly cycles. Generally speaking, this occurs between the ages of 45 and 55, however it can happen at a different time for each person. Hormonal changes, particularly a drop in oestrogen levels, characterise menopause. Women consequently go through a variety of physical and psychological changes, including mood swings, hot flashes, and changes in bone density. Menopause marks the end of women's reproductive potential and the start of a new stage of life in which they are no longer able to conceive naturally. It is distinct from menarche, which denotes the start of the reproductive cycle and involves the first onset of menstruation.

18. Correct option: (C) 23 pairs

The average human cell has 46 chromosomes, which are divided into 23 pairs. Each pair of chromosomes has one from each parent, with one member coming from the mother and the other from the father. These chromosomes include genetic material that affects a variety of features and qualities. The final pair of the 23 pairs consists of sex

chromosomes that determine an individual's biological sex, while the other 22 pairs of the 23 pairs are autosomes responsible for features unrelated to gender. The response "23 pairs" emphasises the significance of these genetic structures in our growth and functioning while also precisely reflecting the number of chromosomes present in a human cell.

18. Correct option: (C) 2

Two sex chromosomes are present in both males and females. The cells of humans contain a total of 46 chromosomes, which are divided into 23 pairs. The majority of qualities are inherited through the use of 22 pairs of these, which are referred to as autosomes. The sex chromosomes in the remaining pair establish a person's biological sex. Males have one X and one Y chromosome, while females have two X chromosomes (XX) (XY). Each person only possesses two sex chromosomes, which are essential for determining their reproductive and sexual traits, regardless of the combination (XX or XY). This response highlights the fact that everyone, regardless of gender, possesses two sex chromosomes.

19. Correct option: (C) Boy, Girl

In humans, the presence of particular sex chromosomes affects how sex is determined. The Y chromosome is inherited from the father, making a child biologically male, whereas the X chromosome is inherited from the mother, making a child biologically female. This is due to the fact that although females have two X chromosomes, males only have one X and one Y chromosome (XY) (XX). The genes that control male development, including the growth of male reproductive organs, are found on the Y chromosome. On the other hand, female development results from the absence of the Y chromosome. As a result, a child who inherits the Y chromosome will grow up to be a boy and a child who inherits the X chromosome will be a girl.

20. Correct option: (A) Proteins

Proteins are necessary for bodybuilding, tissue repair, and maintenance. Amino acids, the building blocks of life, are found in proteins. These amino acids are essential for many biological functions, including the development, maintenance, and repair of tissues like muscles, organs, skin, and bones. Proteins are crucial for muscle growth and recovery after exercise in the context of bodybuilding and fitness. Muscle tissues may micro-tear during physical activity, which needs to be mended in order for muscles to grow. Consuming protein gives the body the essential amino acids to help repair and rebuild muscle fibers, which over time results in more muscular mass and strength. Additionally, proteins serve a variety of other essential bodily roles, such as enzyme

21. Correct option: (D) All the Above

AIDS (acquired immunodeficiency syndrome) is contracted by coming into contact with an infected person through a variety of channels. This covers sexual intercourse, sharing contaminated needles or syringes, and transfusions of blood. The AIDS virus can be spread through blood transfusions using infected blood. The chance of the virus spreading from one person to another when intravenous drug users share needles is very high. Sexual intercourse, particularly unprotected sex with an infected partner, is one of the most popular means of transmission, nevertheless. Human Immunodeficiency Virus (HIV), which causes AIDS, can be found in vaginal secretions, blood, and breast milk. Sexual activity without protection with an infected person so increases the risk of transmission. Therefore, the answer "All the Above" accurately reflects the multiple routes through which AIDS can spread, emphasizing the importance of awareness and prevention measures.

22. Correct option: (B) 28

Ovulation, or the periodic release of eggs from the ovary, occurs in women of reproductive age typically every 28 days. A developed egg is produced from one of the ovaries during ovulation, and it travels down the fallopian tube before potentially being available for fertilization by a sperm cell. Ovulation is a significant occurrence in the menstrual cycle. The average cycle length is 28, though there can be variances depending on the person. Hormonal changes, especially involving the interaction of the hypothalamus, pituitary gland, and ovaries, control this process. Ovulation is a crucial stage in a woman's reproductive cycle since it is when her chances of conceiving are at their maximum.

23. Correct option: (A) Zygote

During fertilization, a zygote is created. A male gamete (sperm) and a female gamete (ova or egg) combine during fertilization to create a zygote. The first cell of a new organism, known as a zygote, carries the genetic makeup of both parents united. The formation of a new person begins with this merger of genetic material. The zygote begins to generate the embryonic features that will eventually mature into a mature organism as it goes through cell division. All of the genetic instructions required to direct the development of numerous tissues, organs, and systems as the embryo grows and matures are present in the zygote.

24. Correct option: (C) Month

It is true that the Latin word "mensis," which meaning "month," is where the word "menstrual" comes from. The phrase is used to describe the cycle of alterations that take place in the female reproductive system on a monthly basis, including the removal of the uterine lining during menstruation. Generally speaking, the menstrual cycle lasts for about 28 days, though there can be variances. The term "menstrual" reflects the cyclical nature of this process, which occurs roughly once a month.

25. Correct option: (C) Sweat and sebaceous

Puberty causes the sebaceous and sweat glands to become increasingly active, which increases perspiration. Significant hormonal changes, principally brought on by the maturing of the reproductive system, take place throughout the puberty stage of development. The sweat glands, often referred to as eccrine glands, are essential for controlling body temperature because they secrete perspiration that evaporates from the skin's surface and cools the body. Sebaceous gland activity also increases during puberty. Sebum, an oily substance produced by these glands, aids in lubricating and protecting the skin and hair. Changes in body odor, acne, and other skin disorders frequently linked to puberty-related hormonal changes might be brought on by the increased activity of the sweat and sebaceous glands.

9. Micro-organisms

Solutions

LEVEL – I

1. Correct option: (A) Curd

Curd, a fermented type of milk, is produced by the action of specific Lactobacillus genus rod-shaped bacteria. These lactic acid bacteria are gram-positive, non-spore-forming bacteria. Lactobacillus bacteria, which metabolize the carbohydrates in milk, produce lactic acid, which drives the conversion of milk to curd.

Lactobacillus is a well-known probiotic component present in foods such as yoghurt and is associated with several health benefits. It assists in the restoration of homeostasis and is especially effective in situations of IBS. The Lactobacillus genus has several distinct species, and curd may include Lactobacillus acidophilus, Lactobacillus lactis, and Lactobacillus lactis cremoris, among others.

2. Correct option: (C) Yeast

Due to its critical function in the fermentation process, yeast, namely Saccharomyces cerevisiae, is added to dough for bread making. When yeast is added to the dough, it metabolizes the carbohydrates present and ferments them into carbon dioxide gas and alcohol. This gas becomes trapped within the dough, allowing it to rise and form air pockets, giving bread its distinctive light and fluffy feel.

Yeast also helps to the flavor development of bread by creating a variety of organic chemicals, including ethanol, and adding a pleasant scent. Yeast is an essential element in bread-making due to its capacity to leaven dough and generate the right texture and flavor.

3. Correct option: (A) Carbon dioxide

Fermentation is a metabolic process performed by microorganisms such as yeast or bacteria that includes the breakdown of organic compounds such as sugars or carbohydrates in the absence of oxygen. This process breaks down complex organic chemicals into simpler compounds such as alcohol or acids, releasing energy in the form of ATP. Carbon dioxide (CO_2) is a particularly noticeable byproduct of fermentation.

This gas is produced by microbes as a byproduct of the breakdown of sugars or carbohydrates and is emitted into the environment as waste. Carbon dioxide generation is widespread in fermentation processes such as alcoholic beverage manufacturing, bread manufacture, and beverage carbonation.

4. Correct option: (C) Alcohol

Yeast is a kind of microbe that plays an important part in fermentation processes. When yeast metabolizes carbohydrates, it goes through a sequence of biochemical events known as alcoholic fermentation. During this process, yeast transforms carbohydrates such as glucose or fructose into alcohol (particularly ethanol) and carbon dioxide. This conversion happens in the absence of oxygen and culminates in the creation of alcohol as the principal end product.

As a result, when carbohydrates are metabolized by yeast, they are transformed to alcohol. Option C) Alcohol is the proper response in this situation.

5. Correct option: (A) Antibiotics

Antibiotics are drugs that are used to treat bacterial illnesses by either killing or inhibiting the development of the bacteria. They accomplish this by specifically targeting certain systems or structures inside the bacterium. Antibiotics efficiently weaken or eradicate bacteria by interfering with critical activities such as cell wall formation or protein biosynthesis. This aids in the cure or treatment of bacterial infections by eradicating the

infection and alleviating related symptoms. Antibiotics are ineffective against viral infections and should only be taken when recommended by a healthcare practitioner to ensure proper and ethical use.
Painkillers (option B) help in managing pain associated with various conditions but do not directly cure the underlying disease or infection. Nutrients (option C) are essential for maintaining overall health and supporting the body's immune system, but they do not cure diseases on their own. Antigens (option D) are substances that stimulate the immune system to produce antibodies, but they are not used to directly cure diseases.

6. Correct option: (D)

7. Correct option: (C) Cow

The name "vaccine" is derived from the Latin phrase "Vaca," which translates to "cow." This relationship is based in the history of vaccination, which was pioneered by Edward Jenner. Jenner discovered that milkmaids who had cowpox were resistant to smallpox.
He later invented a way of inoculating humans against smallpox using material from cowpox lesions, which marked the beginning of vaccination. The name "vaccine" was developed to represent this relationship to cowpox and to recognize the essential role of cows in the creation of this medical achievement.

8. Correct option: (D) Nitrogen

Nitrogen makes up around 78% of the air we breathe. The most abundant gas in the Earth's atmosphere is nitrogen. Oxygen comes in second, accounting for around 21% of the air we breathe.
Nitrogen is a necessary component of life, playing an important part in biological processes and promoting plant development. It also participates in the nitrogen cycle, where it undergoes several changes before being used by organisms. Furthermore, nitrogen helps to maintain atmospheric stability by diluting other gases and inhibiting fast burning.
While other gases such as carbon dioxide, helium, and trace gases account for the remaining 1%, nitrogen dominates our atmosphere and is essential for ecosystem function and the survival of life on Earth.

9. Correct option: (B) Bacteria

Lemon citrus canker is caused by bacteria. Citrus canker is a plant disease caused by the bacterium Xanthomonas citri subsp. citri that damages citrus plants, notably lemons. citri. This bacterium affects citrus trees' leaves, stems, and fruit, causing elevated corky sores on the plant's surface. Wind, rain, and infected tools or equipment are all ways for the bacteria to spread.
Citrus canker can have a substantial economic impact since it affects the market value of afflicted fruits and may lead to tree defoliation and decline. Quarantine procedures, trimming of affected plant portions, and the use of copper-based sprays to suppress the bacteria are all effective management options.

10. Correct option: (B) Spoiled food

Food poisoning is caused mostly by eating damaged or contaminated food. When food is infected with hazardous bacteria, viruses, parasites, or their toxins, it causes food poisoning.
Pathogens can proliferate in food that has not been properly kept, handled, or cooked. Consumption of rotten or contaminated food can cause nausea, vomiting, diarrhea, stomach discomfort, and, in extreme cases, dehydration and organ damage.
To avoid foodborne infections, good food hygiene practices must be followed, including correct food storage, cooking, and handling. Unless infected with hazardous germs, fresh fruits, uncooked food, or poor air quality are unlikely to induce food poisoning.

11. Correct option: (A) 1928

Penicillin, the first widely used antibiotic, was discovered in 1928. It was accidentally discovered by Alexander Fleming, a Scottish bacteriologist, while he was studying bacteria in his laboratory at St. Mary's Hospital in London. Fleming noticed that a mold called Penicillium notatum had contaminated one of his petri dishes, and the bacteria surrounding the mold were being killed.

This observation led to the discovery of penicillin, which revolutionized the field of medicine by effectively treating bacterial infections. The discovery of penicillin laid the foundation for the development of numerous other antibiotics and significantly improved the treatment of infectious diseases.

12. Correct option: (C) Tetracycline

Tetracycline, an antibiotic that has been essential in the fight against bacterial diseases, was discovered by Dr. Y. Subba Rao, who is recognized for this discovery. Dr. Subba Rao initially discovered tetracycline in a soil sample he took in the late 1940s.

This all-purpose antibiotic stops bacterial growth and replication by preventing them from synthesizing proteins. Since its discovery, tetracycline has been widely used to treat a variety of bacterial illnesses, such as skin infections, urinary tract infections, and respiratory tract infections. And also used in medicine and agriculture.

Numerous lives have been saved because to Dr. Subba Rao's ground-breaking research that led to the discovery of tetracycline and set the path for the creation of additional critical antibiotics in the battle against infectious illnesses.

13. Correct option: (B) Pasteur

Louis Pasteur is credited for developing the rabies vaccine. French scientist Louis Pasteur is well known for his contributions to immunology and microbiology. In 1885, he created the first rabies vaccine, which was successfully used to the treatment of a 9-year-old kid who had been bitten by a rabid dog.

In order to weaken the rabies virus, Pasteur first grew it in rabbits and then dried off the affected nerve tissue. It was possible to immunize people with this attenuated virus and elicit an immunological response without actually infecting them.

The creation of the rabies vaccine by Pasteur was a significant advance in the treatment and prevention of this fatal viral illness, saving countless lives and laying the groundwork for contemporary vaccine research.

While Antonie van Leeuwenhoek and Robert Hooke were early pioneers in the area of microscopy, Edward Jenner discovered the smallpox vaccine in 1796. However, nobody of these researchers created a rabies vaccine.

14. Correct option: (C) Jenner

Edward Jenner was an English physician who is credited with discovering the smallpox vaccine in 1796. He observed that milkmaids who had caught cowpox were immune to smallpox, and he developed a vaccine by inoculating people with cowpox virus. The smallpox vaccine was a major breakthrough in public health, and it is credited with saving millions of lives.

Louis Pasteur developed the rabies vaccine in 1885, Alexander Fleming discovered penicillin in 1928, and Jonas Salk developed the polio vaccine in 1955. However, none of these scientists discovered the smallpox virus.

15. Correct option: (C)

The BCG vaccination (Bacillus Calmette-Guérin) protects against TB. The bacteria Mycobacterium bovis, which is closely linked to the bacterium Mycobacterium tuberculosis, which causes tuberculosis, is the source of the live attenuated vaccination known as BCG. In nations where TB is common, the BCG vaccination is often delivered intradermally to newborns soon after delivery.

It encourages the immune system to identify and react to the TB germs, providing some kind of defense against the disease's more severe forms, especially in youngsters. It is crucial to remember that while the BCG vaccination can lower the chance of developing severe forms of TB, it does not offer full immunity from all tuberculosis infections.
MMR, DPT, and ECG are not TB vaccinations. The MMR vaccination protects against rubella, mumps, and measles. The DPT vaccination guards against tetanus, pertussis, and diphtheria. Electrocardiography, sometimes known as an ECG, is a test that captures the electrical activity of the heart.

16. Correct option: (C)

Acquired Immunodeficiency Syndrome, or AIDS, is brought on by the Human Immunodeficiency Virus (HIV). Specifically targeting CD4+ T cells, which are essential for the body's defense against infections and illnesses, HIV is a retrovirus that affects the immune system.
HIV infection causes a person's immune system to progressively deteriorate over time, increasing their vulnerability to different opportunistic infections and certain malignancies. HIV infection can develop into AIDS, an advanced form of the illness marked by significant immune system dysfunction, if untreated.
Unprotected sexual contact, sharing infected needles, and mother-to-child transmission during delivery or nursing are the main ways that HIV is spread. To control HIV infection and halt the development of AIDS, effective antiretroviral medication (ART) is available, enabling people to live longer, healthier lives.

17. Correct option: (C) Yeast

Option C) Yeast is largely responsible for turning sugars into alcohols. An important part of fermentation activities, including the transformation of carbohydrates into alcohol, is played by yeast, a kind of fungus. A process known as alcoholic fermentation is specifically carried out by yeast, in which the enzymes within the yeast cells break down the sugars into simpler chemicals, chiefly ethanol (alcohol) and carbon dioxide.
This method is frequently used to make alcoholic drinks including beer, wine, and spirits. Yeast is the most well-known and frequently used organism for this purpose due to its effectiveness and predictable fermentation capabilities.
While other bacteria and fungi may also contribute to the conversion of carbohydrates into alcohols in specific situations, this is a function that yeast is best recognized and employed for.

18. Correct option: (D) Mosquitoes

Option D) Mosquitoes are the primary carrier of Dengue and Chikungunya. Both illnesses are spread by the bites of infected members of the Aedes genus of mosquitoes, notably Aedes aegypti and Aedes albopictus. These mosquitoes act as carriers of the viruses that lead to Chikungunya and Dengue. The virus is spread by the saliva of the infected mosquito when it bites a human. The virus subsequently multiplies within the body of the affected person, resulting in symptoms including fever, headache, muscular discomfort, and rash. A mosquito that bites a person who has either virus can pick up the virus and then pass it on to other people it bites.
Numerous diseases are spread by mosquitoes, and in the case of Dengue and Chikungunya, reducing the population of mosquitoes and taking precautions against mosquito bites, such as using insect repellents and donning protective clothing, are crucial for halting the spread of these illnesses.

19. Correct option: (D) 1902

Sir Ronald Ross receives the Nobel Prize in Physiology or Medicine. He was awarded this distinguished honor for his discovery of the mosquito-borne transmission of malaria, particularly the function of Anopheles mosquitoes in this process.

This discovery made a substantial contribution to our comprehension of the malaria transmission cycle and opened the door for further investigation and treatment options for this deadly illness.

20. Correct option: (B) Groundnut

A fungus called tikka disease harms groundnut plants. The fungi Cercospora personata and Cercospora arachidicola are the culprits. Brown patches that form on the groundnut plant's leaves are the disease's primary symptom. These blotches may ultimately combine to produce substantial lesions, which may cause the plant to lose its leaves.

For farmers of groundnuts, tikka disease can be a big issue since it drastically reduces production. The illness may be spread by wind, rain, and insects and is most prevalent in warm, humid settings.

There are several methods for managing the tikka illness, including:

planting resistant groundnut varieties

crops in rotation

burning plant trash with a disease

Fungicide use.

LEVEL-II

1. Correct option: (A) Vibrio cholerae is an example of type S, which causes cholera

Based on their morphologies, bacteria are divided into four main types:

- Spherical (cocci): These microorganisms have an oval or spherical form. E.g., Streptococcus pyogenes.
- Bacilli: These bacteria have a rod form. E.g., Lactobacillus
- Spirilla: These bacteria have a spiral form. Consider the Spirillum minus.
- Vibrios: These bacteria have comma shapes. Eg. Vibrio cholerae.

In the given options correct match is A) Vibrio cholerae is an example of type S, which causes cholera. The bacteria that cause cholera is called Vibrio cholerae. When left untreated, cholera, an acute diarrheal illness, can be fatal and cause severe dehydration. Especially in places with inadequate sanitation and no access to clean water, Vibrio cholerae is primarily spread by contaminated food or water.

The bacteria secrete a toxin that damages the intestines and causes dehydration, vomiting, and the typical cholera symptoms of copious watery diarrhea. To manage cholera patients and stop its spread in impacted populations, quick rehydration and antibiotic therapy are essential

2. Correct option: (A) X - Streptococcus; Y - Aerobic respiration

Yeast from the Saccharomyces genus is frequently employed in the manufacture of wine and alcohol for commercial purposes. It is renowned for its capacity to effectively carry out the fermentation process that turns sugar into alcohol. Brewer's yeast or baker's yeast, sometimes referred to as Saccharomyces cerevisiae, is a well-known species in this genus and is frequently employed in the fermentation of alcoholic beverages.

Through the activity of microorganisms like yeast, fermentation is the process of turning carbohydrates, such as glucose or fructose, into alcohol and carbon dioxide. Alcohol (ethanol) is a result of the yeast's metabolism of the carbohydrates during fermentation. This method is frequently used to make alcoholic drinks as well as other products, such vinegar that contains acetic acid.

3. Correct option: (D) It is a parasitic fungus that causes various diseases in plants, animals, and humans

A filamentous fungus called Rhizopus stolonifer, sometimes referred to as black bread mould, is found on bread surfaces. It gets its nutrition from the bread, which leads to deterioration and destruction. As an opportunistic pathogen, this global mould may

potentially cause zygomycosis infections in people with compromised immune systems. It may result in cutaneous, lung, or sinus infections in certain situations.
Rhizopus stolonifer is also known to produce soft rot in sweet potatoes, which results in dark, squishy regions and an unpleasant odor by decomposing the tissues of the vegetable. In order to avoid contamination and reduce the hazards brought on by Rhizopus stolonifer and other moulds, it is crucial to handle food properly and practice good hygiene.

4. Correct option: (C) A – iv, B – ii, C – i, D – iii

A. Varicella zostera (virus) - iv. Chickenpox
The extremely infectious viral ailment known as chickenpox, which results in painful, blister-like skin rashes, is brought on by the varicella zoster virus.
B. Bordetella pertussis - ii. Whooping cough
Whooping cough is a highly infectious respiratory infection caused by the bacteria Bordetella pertussis, which is characterized by violent coughing fits and a unique "whooping" sound when inhaled.
C. Rhinovirus - i. Common cold
One of the major viruses to blame for the common cold, a moderate viral infection of the nose and throat that frequently includes symptoms like runny nose, sneezing, and congestion, is the rhinovirus.
D. Rubeola virus - iii. Measles
Measles is a highly infectious viral disease that predominantly affects children and is brought on by the Rubeola virus. A red rash, high fever, cough, runny nose, and other flu-like symptoms are its defining features.

5. Correct option: (B) Microorganisms exist in soil, water, air, food, etc.

'Omnipresent' refers to being widely dispersed or present everywhere. In all surroundings, including soil, water, and the air, as well as inside the bodies of plants, animals, and people, microorganisms may be found. By taking part in nutrient cycling, decomposition, and other ecological processes, they perform crucial roles in ecosystems. The term "omnipresent" is thus supported by option B since it accurately depicts the broad existence of microbes in many settings.

6. Correct option: (D) Only 1

Fungi – Eukaryotic, unicellular, or multicellular
Fungi are eukaryotic organisms that can exist as either unicellular (yeast) or multicellular (molds and mushrooms). This statement is correct.
Bacteria – Eukaryotic, multicellular
This statement is incorrect. Bacteria are prokaryotic organisms, meaning they lack a true nucleus and other membrane-bound organelles. They are typically unicellular.
Protozoa - Prokaryotic, parasite
This statement is incorrect. Protozoa are eukaryotic organisms and not prokaryotic. They are unicellular and often free-living, although some can be parasitic.
Virus – Eukaryotic, multicellular
This statement is incorrect. Viruses are acellular particles that are not considered living organisms. They are composed of genetic material (DNA or RNA) enclosed in a protein coat and cannot replicate on their own. They are not eukaryotic or multicellular.

7. Correct option: (D) 1,2 & 3

In reaction to an outside threat, such as a virus or bacterium, the immune system of the body produces proteins called antibodies. They are made to recognize and bind to the foreign material with particularity, which aids the immune system in destroying it.
The first claim is that antibodies are chemical defenses that battle infections. This is true because antibodies are proteins made particularly to attach to and recognize alien objects

like viruses and bacteria. This binding aids in the immune system's ability to eliminate the foreign material.
Statement 2: Proteins make up antibodies. Additionally, since antibodies are proteins, this is true. They are composed of amino acids and are created by B cells in the immune system.
Statement 3: By eliminating the pathogens, antibodies save us from sickness. This is also true since antibodies may assist the immune system via a number of techniques in destroying foreign substances.

8. Correct option: (A) Both A and R are true, and R is the correct explanation for A

The baking sector utilizes yeasts like Saccharomyces cerevisiae because they create carbon dioxide during fermentation. Through thermal expansion, this carbon dioxide makes the bread dough rise. The cause is that when the dough heats up, the stored carbon dioxide gas expands and causes the dough to rise.
As a result, the reason is the proper justification for the statement, and both the assertion and the reason are true.
Assertion: Yeasts such as Saccharomyces cerevisiae are used in baking industry.
Yeasts are single-celled fungi that can ferment carbohydrates to release alcohol and carbon dioxide. Bread dough rises as a result of this fermentation process. Because it is so effective at creating carbon dioxide, the yeast Saccharomyces cerevisiae is frequently employed in baking.
Reason: Carbon dioxide produced during fermentation causes bread dough to rise by thermal expansion.
When yeast ferments sugars, it produces carbon dioxide gas. This gas is trapped in the bread dough, and as the dough heats up in the oven, the gas expands, causing the dough to rise. The thermal expansion of the carbon dioxide gas is what causes the bread to become light and fluffy.

9. Correct option: (C) Both the statement is correct

Statement 1: Vaccines are the weakened disease-causing microorganisms that are injected into our body to develop immunity.
This claim emphasizes the idea that disease-causing bacteria are used in vaccinations in weakened or inactivated forms. In order to trigger an immune response, infections or their components are frequently weakened, destroyed, or injected into the body as part of vaccines. The immune system can learn to identify and create a defense against the genuine disease-causing germs by being exposed to these weaker forms.
Statement 2: Vaccination is the administration of antigenic material to stimulate the immune system of a person to protect against diseases.
In this sentence, the goal and broader definition of vaccination are emphasized. Antigenic material, which can comprise pathogens that have been weakened or rendered inactive as well as certain proteins or genetic components originating from the pathogens, is administered during vaccination. The objective of the immune system stimulation and immunological response is to protect the individual receiving the vaccination from further infections by the disease-causing germs.
In conclusion, Statement 2 offers a broader perspective by emphasizing that vaccination involves the administration of antigenic material to stimulate the immune system and provide protection against diseases, whereas Statement 1 concentrates on the use of weakened disease-causing microorganisms in vaccines.

10. Correct option: (A) Rhizobium bacteria

Rhizobium bacteria are those that support the transformation of atmospheric nitrogen into nitrite or nitrate in conjunction with leguminous plants. These bacteria live in nodules on the roots of leguminous plants in a symbiotic relationship.

Rhizobium bacteria use a process known as nitrogen fixation to transform atmospheric nitrogen into a form that plants may use for growth and development. For the purpose of supplying the soil with nitrogen and enhancing plant health, the mutualistic interaction between Rhizobium bacteria and leguminous plants is crucial.

11. Correct option: (C) Virus

A virus is a type of microbe that can only multiply inside the cells of its host. The unique infectious agent known as a virus needs a host cell to replicate and grow. Outside of a host cell, they are unable to proliferate independently. Once a virus has entered a host cell, it exploits the cellular machinery to manufacture new virus particles, which eventually burst out of the host cell, harming the cell and dispersing the infection. Viruses have a unique mechanism of reproduction that sets them apart from other microbes like diatoms, flagella, and Rhizopus.

12. Correct option: (C) Malaria & Amoebiasis

Malaria and amoebiasis are two of the diseases produced by protozoa.

The protozoan parasite of the genus Plasmodium, which causes malaria, infects female Anopheles mosquitoes and spreads to people by their bites.

Entamoeba histolytica is a protozoan parasite that causes amoebiasis, often known as amoebic dysentery. It typically spreads when contaminated food or water is consumed.

The other diseases mentioned in the options have different causative agents:

- Smallpox is caused by the variola virus.
- Tuberculosis is caused by the bacterium Mycobacterium tuberculosis.
- Typhoid is caused by the bacterium Salmonella typhi.
- Chikungunya is caused by the chikungunya virus.

13. Correct option: (D)

Since most microorganisms are so tiny, they cannot be seen with the human eye. They generally range in size from a few micrometers to submicron or nanometer scale, making them tiny. Only a microscope will allow you to see them.

Both "microorganisms are helpful" and "microorganisms are dangerous" are true. Depending on the species, microorganisms can either be advantageous or harmful. For instance, while certain microbes are capable of causing disease, others can aid in the digestion of food and the production of vital nutrients.

Therefore, "microorganisms can be seen by the naked eye" is the sole claim that is false.

14. Correct option: (D) Statement 3 is correct only

Statement 1: Fishes are preserved by adding sugar.

This is incorrect. Fish can be preserved via smoking, drying, or salting.

Sugar draws moisture, which makes it a bad preservative for fish since it might inducerott ing.

Statement 2: Jams and jellies are preserved by adding salt.

Sugar is used to preserve jams and jellies.

Jams and jellies shouldn't be preserved with salt since it could impart a salty flavor.

Statement 3: Vegetable slices are preserved by adding salt and drying in sunlight.

This is correct. By salting vegetable slices and letting them dry in the sun, you may preserve them. Vegetables are dried out by the salt, which helps keep them from rotting. Any microbes that might be on the veggies are also assisted by the sunshine in their destruction.

15. Correct option: (C) W – Virus, X – Fungus, Y – Protozoan

The microorganisms are classified based on their characteristic structures

W – Virus - The cells of the host organism are the sole area where viruses may reproduce. Viruses are contagious, non-living creatures. They are composed of genetic material (DNA or RNA) covered with proteins called a capsid, which is composed of tiny units called capsomeres. These capsomeres are structured geometrically in helical or polyhedral shapes.

X – Fungus - Eukaryotic creatures called fungi take nutrients from their environment by absorbing them. They may reproduce through spores and are distinguished by the filamentous structures they possess, known as hyphae.
Y – Protozoan - Eukaryotic single-celled creatures called protozoa exist. They fall under the protist kingdom classification and are found in a variety of habitats. The different forms and means of protozoa's motility, such as flagella, cilia, or pseudopodia, are well recognized.

16. Correct option: (A) P – Air, Q – Tuberculosis, R – Bacteria, S - Malaria

P is spread by the air, and a virus is to blame for the disease. Viral illnesses like measles and mumps can be transmitted by respiratory droplets.
Q is the tuberculosis disease which is spread by the air, and the bacterium Mycobacterium tuberculosis that causes airborne illness TB.
R is bacteria Salmonella typhi that may spread by tainted food and water that causes typhoid fever.
S is the malaria disease spread by mosquitoes, which act as vectors, and a protozoan causes the disease. The protozoan parasite of the genus Plasmodium is the root cause behind malaria, an illness spread by mosquitoes.

17. Correct option: (B) It is a type of virus which attacks bacteria

A bacteriophage, sometimes referred to as a phage, is a kind of virus that infects and spreads throughout bacteria. It targets bacteria selectively and reproduces by using them as host cells. Bacteriophages are multi-layered organisms that contain genetic material (DNA or RNA) that is covered in a protein coat.
In order to make additional phages, they adhere to the surface of bacterial cells, inject their genetic material there, and take over the host's cellular machinery. Bacteriophages eventually cause infected bacterial cells to lyse (break open), which releases fresh phages that can infect further bacterial cells.

18. Correct option: (D) 1, 2 & 3

To avoid food poisoning, food must be kept properly: The growth of bacteria and other pathogens that might result in food poisoning is slowed down by proper food storage practices, such as refrigeration at the right temperatures.
Longer periods of time without refrigeration can cause food poisoning: Bacteria may quickly develop when perishable foods are kept at room temperature for a lengthy period of time, increasing the risk of foodborne diseases.
Food handling errors can also result in food poisoning. For example, improper hand hygiene, mixing raw and cooked meals, and using contaminated utensils or surfaces can all lead to the introduction of bacteria into the food.

19. Correct option: (C) b only

Autotrophs are creatures that have the ability to synthesize their own food via chemosynthesis or photosynthesis. Algae are photosynthesis-capable creatures that can absorb solar energy and use photosynthesis to transform it into chemical energy. They make use of this energy to create organic compounds like carbohydrates, which are their main food supply.
An oxygen-free kind of respiration is called anaerobic respiration. Algae do not require anaerobic respiration since they do not utilize it.
Heterotrophic feeding occurs when an organism consumes another live thing as well as dead ones. Since algae are autotrophs, they don't require a source of food that is either living or dead.

20. Correct option: (B) (i) - (e); (ii) - (c); (iii) - (a); (iv) - (b); (v) - (d)

i. Louis Pasteur made significant contributions to the field of microbiology and fermentation. In 1857, he conducted experiments that demonstrated the role of microorganisms in the process of fermentation.

ii. Alexander Fleming discovered the antibiotic substance known as penicillin in 1929. His discovery revolutionized the field of medicine by introducing the first widely used antibiotic.
iii. Edward Jenner is credited with developing the first successful smallpox vaccine in 1798. His work laid the foundation for the field of immunization.
iv. Robert Koch was a renowned microbiologist who identified the bacterium Bacillus anthracis as the causative agent of anthrax in 1876.
V Anton van Leeuwenhoek was a Dutch scientist who is considered the father of microbiology. He was the first to observe and describe microorganisms using a simple microscope, which he designed and built himself

21. Correct option: (A) Only 2

Statement 2 states that Chikungunya is an infectious disease caused by bacteria. However, this statement is incorrect. The Chikungunya virus, which is a member of the Togaviridae family, is what really causes the disease.
Chikungunya is spread to people through the bite of infected mosquitoes, especially Aedes mosquitoes, which are also known to spread other viral illnesses including dengue and Zika. When the Chikungunya virus infects someone, they may get symptoms including a high temperature, joint discomfort, rash, headache, and muscle soreness.

22. Correct option: (D) A, B & D

A) A single-celled fungus called yeast is utilized in fermentation. Alcohol and carbon dioxide are produced from sugar during the fermentation process. Bread, beer, wine, and other foods that go through the fermentation process all utilize yeast. Therefore, it is a correct match.
Bacteria that can fix nitrogen include blue-green algae. The process of nitrogen fixation involves transforming atmospheric nitrogen into a form that plants can utilize. Due to their ability to enhance the quantity of nitrogen in the soil, blue-green algae are crucial for soil fertility. Therefore, it is a correct match.
A bacteria called Rhizobium collaborates symbiotically with legumes. In this interaction, a bacterium that fixes nitrogen resides in the root nodules of legume plants. The legume plant then has access to this nitrogen, aiding in the improvement of the plant's growth. Therefore, it is an incorrect match
The fungus penicillium is responsible for making penicillin. Several different bacterial infections can be treated with penicillin. Therefore, it is a correct match.

23. Correct option: (B) C, D, A and B

C)Dig a pit - This is the initial stage of the manure-making process. The pit needs to be large enough to hold all of the organic waste that will be added, and it needs to be excavated in a well-drained region.
D)Dump the organic waste into pit - Food scraps, animal dung, and other organic elements are examples of this. To help keep the waste wet and stop it from luring pests, it should be covered with a layer of dirt.
A) Action of microorganisms - Microorganisms will eventually decompose the organic waste. The waste will be transformed into manure by these microbes, which is a great source of nutrients for plants.
B) Formation of manure - The manure will be ready to use after a few months. It can be applied to gardens and lawns to improve soil fertility.

24. Correct option: (A) a, c and d

a) Antibiotics are manufactured from fungi and bacteria: In 1928, Alexander Fleming isolated the first antibiotic, penicillin, from a fungus known as Penicillium notatum. Since then, several additional antibiotics derived from fungus and bacteria have been found.
b) Antibiotics are synthesized in our body: Our bodies cannot produce antibiotics. They are made by other living things like fungus and bacteria.

c) Antibiotics are not effective against viral diseases: Bacteria are either killed or have their growth slowed by antibiotics. As viruses are a distinct kind of creature, they are ineffective against them.
d) Antibiotics are mixed with the feed of livestock: In order to stop the transmission of illnesses among animals, this is done. Antibiotic resistance in bacteria has, however, been related to the use of antibiotics in cattle feed.

25. Correct option: (C) D, B, A and C

1. Food spoilage involves breakdown of food by toxic enzymes. This is the first step in the process of food spoilage. The enzymes are produced by microorganisms, such as bacteria and fungi, that are present in the food.
2. Evolution of carbon dioxide. As the food spoils, the microorganisms will produce carbon dioxide. This carbon dioxide will cause the food to become bloated and mushy.
3. Food becomes stale, smelly, mushy. The food will also become stale, smelly, and mushy as it spoils. This is due to the breakdown of the food by the microorganisms.
4. Stale food causes illness. If stale food is eaten, it can cause illness. This is because the microorganisms that have spoiled the food can also produce toxins that can make people sick.

26. Correct option: (A) Both A and R are true, and R is the correct explanation for A.

In fact, the symbiotic relationship between algae and fungus produces lichens.
While the fungus creates a protective framework and may absorb water and nutrients, the algae contribute photosynthetic powers and food production for the partnership.
Lichens may exist and flourish in a variety of habitats thanks to their mutualistic interacti on.
The explanation (R), which emphasises the distinct functions that both algae and fungus play in this interaction, also adequately explains why lichens include a symbiotic associati on.
The proper response is therefore A) Both A and R are true, and R is the appropriate justifi cation for A.

27. Correct option: (A) (i) → (e); (ii) → (a); (iii) → (c); (iv) → (b); (v) → (d)

(i) Canning - (e) Sealed food in metal: Food is preserved via canning, which involves sealing it in airtight vessels, usually made of metal. The longer the food is preserved, the more protection against air and bacteria it receives from this procedure.
(ii) Chemical preservative - (a) Vinegar: Chemical preservatives are compounds that are applied to food to stop microbial development and lengthen shelf life. Due to its antibacterial qualities, vinegar, which is acidic, functions as a natural chemical preservative.
(iii) Pasteurization - (c) Heat and cold treatment: In order to pasteurize food or drink, it must first be heated to a certain temperature and then quickly cooled. It also helps to prolong the shelf life of the items by killing or inactivating dangerous microorganisms.
(iv) Pickling - (b) Salt and oil: Food is submerged in a solution containing salt, vinegar, or brine as part of the pickling preservation technique. This method aids in establishing conditions that prevent bacterial development and increase the food's shelf life. Pickling can also use oil for flavor and further preservation.
(v) Freeze drying - (d) Froze ice to vapor: Freeze drying, sometimes referred to as lyophilization, is a method of food preservation that entails freezing the food and then sublimating the ice to remove it. This method considerably increases the food's shelf life while maintaining the food's natural flavor, texture, and nutritional content.

28. Correct option: (C) III and IV

Statement I: is correct: An individual's immunity is their capacity to fend off illnesses. This is accomplished through the immune system's coordinated reaction to any foreign particles.

Statement II: is correct: The formation of antibodies, the activation of white blood cells, and the release of chemicals that might destroy or weaken foreign particles are just a few of the processes that may be involved in this response.
Statement III: is incorrect: Cyanobacteria are the bacteria that encourage curd production.
Actually, Lactobacillus, not Cyanobacteria, is the bacterium that aids in the production of curd.
Photosynthetic microorganisms called cyanobacteria are typically found in watery habitats.
Statement IV: is incorrect: Spirogyra sp. is a type of algae, not a fungus. Algae are photosynthetic organisms that are found in both freshwater and saltwater environments. Fungi are heterotrophic organisms that decompose dead organic matter.

29. Correct option: (A) X – Bacteria, Y – Fungi, Z – Virus

Citrus canker: A bacterial condition known as citrus canker is brought on by Xanthomonas axonopodis pv. citri. The bacteria can spread through the water, the air, or plant matter that has been contaminated.
Rust of wheat: Puccinia graminis is the fungus that causes wheat rust. The fungus can spread through the air or on plant material that has been contaminated.
Yellow vein mosaic of bhindi: The cucumber mosaic virus is the cause of the viral illness known as yellow vein mosaic of bhindi. Infected seeds, mechanical devices, or insects can all spread the virus.

30. Correct option: (C) Only B

Due to their capacity to prevent bacterial development, oil and vinegar are frequently employed as preservatives. Food preservation is made possible by the fact that both oil and vinegar foster an environment that is unfavourable for bacterial development. While oil serves as a barrier to keep germs from reaching the food and reducing oxygen exposure, the acidity of vinegar can produce an unfavorable habitat for some bacteria.

10. Cell-The Basic Unit of Life

Solutions

LEVEL-I

1. Correct option: (A) Oak tree

Cork has dead cells which resembles the cells found in Matchstick. Onion peel cells and Cheek cells are living cells.

2. Correct option: (B) Robert Brown

Robert Brown discovered nucleus in 1831.

3. Correct option: (D) Nucleus

Robert Brown observed nucleus in a plant cell while studying the process of fertilization and claimed that it is an integral part of any cell, in the year 1831.

4. Correct option: (A) Orchid

Robert Brown coined the term 'nucleus' in 1831. He first observed the nucleus in the epidermis of the orchids as an opaque spot.

5. Correct option: (A) Cytoplasm

Cytoplasm is the jelly-like substance present between the cell membrane and the nucleus. It is mainly composed of water, salts, and proteins. All the membrane bound cell organelles like endoplasmic reticulum, golgi complex, etc are present in the cytoplasm.

6. Correct option: (D) All of the above

Unicellular organisms include all prokaryotes including Bacteria, and Eukaryotes including Chlamydomonas sp, and Amoeba sp.

7. Correct option: (C) Amoeba sp.

Pseudopodia or false feet are cytoplasmic extensions present in unicellular protists, and functions in locomotion and capturing of prey.

8. Correct option: (A) Protection

The dermal tissue covers the outer surface of the stem and usually functions to waterproof, protect, and control gas exchange.

9. Correct option: (D) All of these

Any process that is commonly exhibited by living organisms on a daily basis for carrying out daily metabolic processes can be termed as a life process.

10. Correct option: (C) Janssen

A Dutch father-son team named Hans and Zacharias Janssen invented the first so-called compound microscope in the late 16th century when they discovered that, if they put a lens at the top and bottom of a tube and looked through it, objects on the other end became magnified.

11. Correct option: (D) Cell

The cell is the basic structural and functional unit of life. Cells are independent, single-celled organisms that take in nutrients, excrete wastes, detect and respond to their environment, move, breathe, grow, and reproduce.

12. Correct option: (C) Microscope

Microscope, instrument that produces enlarged images of small objects, allowing the observer an exceedingly close view of minute structures at a scale convenient for examination and analysis.

13. Correct option: (D) Sclerenchyma cells

Mature sclerenchyma cells are usually dead cells that have heavily thickened secondary walls containing lignin. The cells are rigid and non-stretchable and are usually found in nongrowing regions of plant bodies, such as the bark or mature stems.

14. Correct option: (A) The number of cells present
The size of an organism primarily depends upon the number of cells in its body. When the number of cells increases then the size of the organism also increases.
15. Correct option: (A) Cell wall
The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress.
16. Correct option: (D) Both A and B
The functions of pseudopodia include locomotion and ingestion: Pseudopodia are critical in sensing targets which can then be engulfed; the engulfing pseudopodia are called phagocytosis pseudopodia.
17. Correct option: (A) 1-A; 2-D; 3-B; 4-C
Robert Hooke in the year 1665 discovered cell in cork tissue. Robert brown discovered nucleus in orchids in the year 1831. Antonie van Leuwenhoek made the first microscope and visualized live cells. A plant cell is a living cell due to a proper functioning metabolism.
18. Correct option: (D) 2 only
Robert Brown was the one who discovered nucleus while studying orchid leaves and coined the term 'nucleus' in 1831.
19. Correct option: (D) Both A and B
The shape of the human red blood cell is known to be a biconcave disk.
20. Correct option: (D) 90-100 cm
The nerve cell is the structural and functional unit of the nervous system. They are generally long and branched cells and can be 90 - 100 cm long.

LEVEL II

1. Correct option: (B) Cytoplasm
The cytoplasm is a thick, clear, jelly-like substance present inside the cell membrane. Most of the chemical reactions within a cell take place in this cytoplasm. The cell organelles such as endoplasmic reticulum, vacuoles, mitochondria, ribosomes, are suspended in this cytoplasm.
2. Correct option: (D) Cell wall, Chloroplast, Mitochondria, Cell membrane, Nucleus
The following are the components of animal cells: Nucleus, Ribosomes, Endoplasmic reticulum, Lysosomes, Golgi bodies, Cell membrane, Cell wall, nuclear membrane, Mitochondria.
3. Correct option: (C) c and d
As red blood cells contain no nucleus, protein biosynthesis is currently assumed to be absent in these cells. Since its an animal cell it doesn't have a plastid.
4. Correct option: (C) b and d
A nerve cell or neuron is so long about 100 cm so that it can carry messages over long distances from brain to other parts of the body. It has numerous projections to make contact with other neurons.
5. Correct option: (B) Euglena, Chlorella, Escherichia
Euglena is a unicellular protist, Chlorella is an unicellular algae growing in fresh water. Escherichia coli is a gram-negative bacterium hence an unicellular organism.
6. Correct option: (C) A-ii, B-iv, C-I, D-iii
A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.
7. Correct option: (A) Mitochondria - synthesize the respiratory enzymes
Mitochondrion is the site for aerobic respiration in cells. It contains respiratory enzymes necessary for ATP synthesis. A nucleus is the membrane-enclosed organelle within a cell that contains the chromosomes. Cytoplasm is the gelatinous liquid that fills the inside of a

cell. It is composed of water, salts, and various organic molecules. A cell wall is defined as a rigid, external layer that is specifically designed to provide structural support and rigidity.

8. Correct option: (B) X – Plastid, Y – Chloroplast, Z – Chromoplast

The chloroplast is the photosynthetic plastid, and it is named for its green colour. Based on pigmentation, the non-photosynthetic plastids can be broadly divided into leucoplasts, the 'white' or colourless plastids, and chromoplasts, the coloured plastids notable for their accumulation of carotenoids.

9. Correct option: (A) a and b only

Nucleus is the largest cell organelle and controls the activities of the cell. The living material of a cell that is made up of both nucleus and cytoplasm is called protoplasm. Chloroplasts are only one, albeit the most prominent, member of a larger family of plant organelles called plastids. All plastids contain the same genome as chloroplasts, but they differ in both structure and function.

10. Correct option: (D) iii and iv only

Plant cells have a cell wall, chloroplasts and other specialized plastids, and a large central vacuole, whereas animal cells do not.

11. Correct option: (A) These are composed of tiny spherical structures surrounded by a double membrane

Lysosomes are solitary membrane-bound organelles that contain digestive enzymes that break down food scraps, bacteria, macromolecules, worn-out organelles, and viruses that have been eaten. Lysosomes do not have a double membrane. They are also called suicide bags of cell as they can digest their own cell.

12. Correct option: (A) i, iii and iv only

Anything inside the plasma membrane is called a protoplasm. This term was firstly introduced by J. E. Purkinje. Rudolf Virchow gave the statement *'Omnis cellula-e-cellula'*. It means "All cells arise from pre-existing cells. The Cell Theory was first given by M.J Schleiden and Theodor Schwann in 1839.

13. Correct option: (B) The egg cell has half as many chromosomes as the skin cell

Egg and sperm cells have just 23 chromosomes each. That's half as many chromosomes as regular cells. Through the process of fertilization, egg and sperm join to make a cell with 46 chromosomes (23 pairs), called a zygote.

14. Correct option: (D) A pancreatic cell engaged in the production of digestive enzymes

Endoplasmic reticulum helps in the transport of substances within the cell. Rough endoplasmic reticulum (RER) bears ribosomes on the cytoplasmic surface and synthesises secretory and membranous proteins. Then the modified proteins move into the transitional area where they are further processed and packed into secretory vesicles. Thus, RER is rich in cells which are actively engaged in protein synthesis and secretory activity, e.g., pancreatic cells engaged in the production of digestive enzymes, etc.

15. Correct option: (B) X could be Amoeba whereas Y could be Mycoplasma

An amoeba is an eukaryotic unicellular cell, that have a well-organized nucleus which is separated from the cytoplasm by the nuclear membrane. Mycoplasmas are the smallest bacterial cells that do not have a true nucleus and their nuclear material is in direct contact with cytoplasm, also they do not have a cell wall.

16. Correct option: (D) Leucoplasts help in the storage of Food

Plastid is the organelle that found only in plant cells. The three types of plastids are green plastid, nongreen chromoplast, and leucoplasts. The leucoplasts are the non-colored plastid which is used for the storage of the food. The oil, starch, and fats are generally stored in these plastids.

17. Correct option: (C) Only D

Chloroplast are involved in synthesis of food for plant through photosynthesis. Nucleus is a central organelle responsible for controlling all metabolic reactions and contains genetic material of the cell. The cell wall surrounds the plasma membrane of plant cells and provides tensile strength and protection against mechanical and osmotic stress. Mitochondria is involved in production of energy in the form of ATP.

18. Correct option: (D) Only II

A vacuole is a membrane-bound cell organelle. In animal cells, vacuoles are generally small and help sequester waste products. In plant cells, vacuoles help maintain water balance. Sometimes a single vacuole can take up most of the interior space of the plant cell.

19. Correct option: (C) A is true but R is false

The endoplasmic reticulum which lacks ribosomes is called smooth endoplasmic reticulum and is involved in lipid biosynthesis.

20. Correct option: (A) Golgi complex

Golgi complex receives the substances synthesized and release by the ER, condenses, modifies, packs and releases them in the form of secretory vesicles.

21. Correct option: (C) A – Incoming transport vesicle, B – Lumen, C - Newly form vesicle, D – Cisternae, E –Secretory vesicle

Golgi apparatus also called Golgi complex and found universally in both plant and animal cell. It is comprised of a series of membrane-covered sacs called cisternae that look something like a stack. The cis face of a Golgi stack is also known as forming a face which has a convex face. The trans face is known as the maturing face which has a concave face. From the forming face, several vesicles are pinched off forming.

22. Correct option: (C) Both the statements are correct

Antonie van Leeuwenhoek was the first to see living bodies under a Microscope in the year 1674. Robert Brown discovered the nucleus in the Orchid leaf in the year 1831.

23. Correct option: (A) D, C, B, A

Take a scaly leaf of onion and pull a transparent peel. Place the peel onto a drop of water on a glass slide. Add a drop of methylene blue and place a coverslip. Observe the slide under a microscope.

24. Correct option: (D) A – v, B – iii, C – i, D – ii, E – iv

Onion cell is plant cell that have a cell wall surrounding its membrane. Chromosomes are strands of DNA interwound together and forms a distinct mass during cell division. Methylene blue is a bluish coloured stain. Microscope is an instrument that magnifies and aids in visualization of tiny creatures. Ribosomes are protein manufacturing organelles that maybe freely suspended in cytoplasm or maybe attached on the membrane of rough endoplasmic membrane.

25. Correct option: (D) i, ii, and iii

In 1972, Singer and Nicolson proposed the fluid mosaic model of the plasma membrane. The fluid mosaic model describes the plasma membrane structure as a mosaic of components -including phospholipids, cholesterol, proteins, and carbohydrates. The proportions of proteins, lipids, and carbohydrates in the plasma membrane vary with the cell type.

26. Correct option: (A) Both A and R are true, and R is the correct explanation of A

Chloroplasts are the green plastids. They contain a green coloured pigment called chlorophyll. Chlorophyll is the key component in the process of photosynthesis which is the preparation of food for the plant. Therefore, chloroplast is known as the kitchen of the cell.

27. Correct option: (A) Vacuoles

Vacuoles are membrane-bound organelles found in the cytoplasm of eukaryotic cells. They can vary in size, but they are typically large, bulged, and transparent structures that play various essential roles in the cell. One of their primary functions is to store water and maintain cell turgidity. Vacuoles can also store waste products, excretory substances, and even toxic materials in some cases.

28. Correct option: (D) a, b, and c

Chromosomes are present inside nucleus and are seen only during cell division and carries genes on it.

29. Correct option: (D) Cell Wall

Cell wall is a protective outermost layer, which is present surrounding the cell membrane. It is present in plant cells. It supports the contents of cell and protects from pathogen attack and wear and tear. It prevents water loss from plant cells. It gives shape to the plant. It is made up of cellulose and lignin.

30. Correct option: ** A-Thylakoid, B- Granum, C- Outer Membrane, D-Intermembranous space, E- Inner Membrane

Chloroplasts are found in all higher plants. It is oval or biconvex, found within the mesophyll of the plant cell. The size of the chloroplast usually varies between 4-6 μm in diameter and 1-3 μm in thickness. They are double-membrane organelle with the presence of outer, inner and intermembrane space. There are two distinct regions present inside a chloroplast known as the grana and stroma. Grana are made up of stacks of disc-shaped structures known as thylakoids or lamellae. The grana of the chloroplast consists of chlorophyll pigments and are the functional units of chloroplasts.